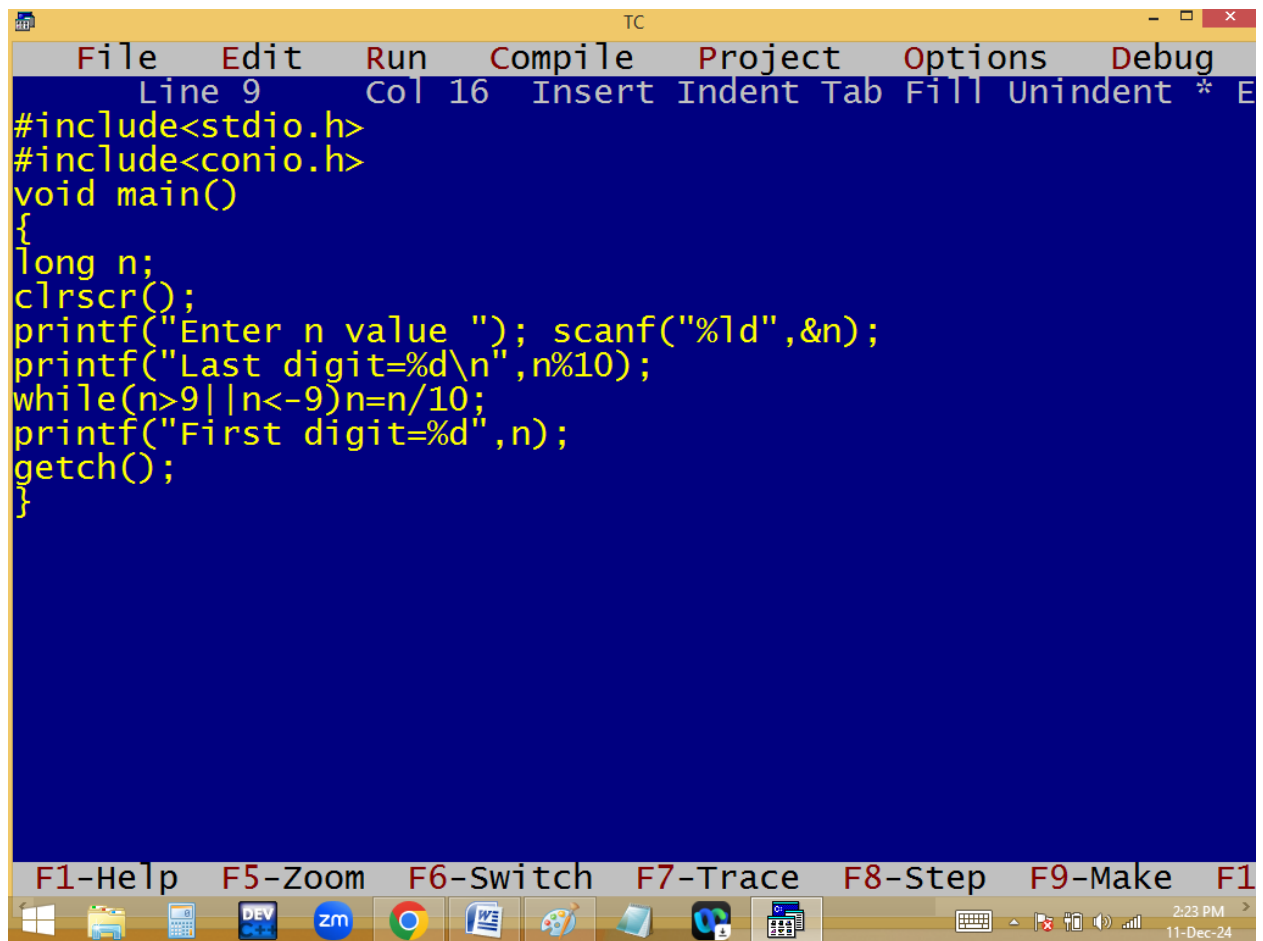


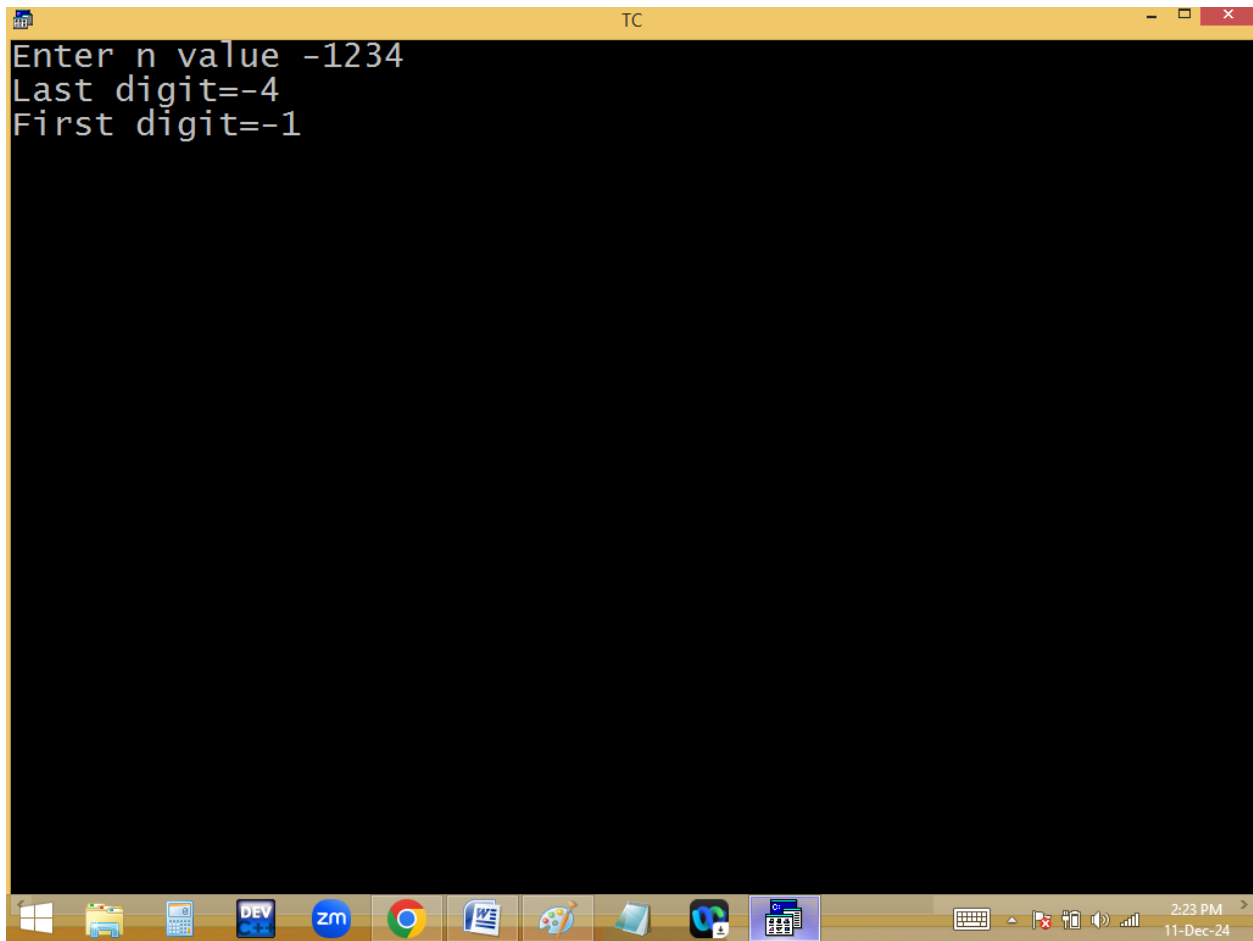
Printing 1st and last digit of given no.

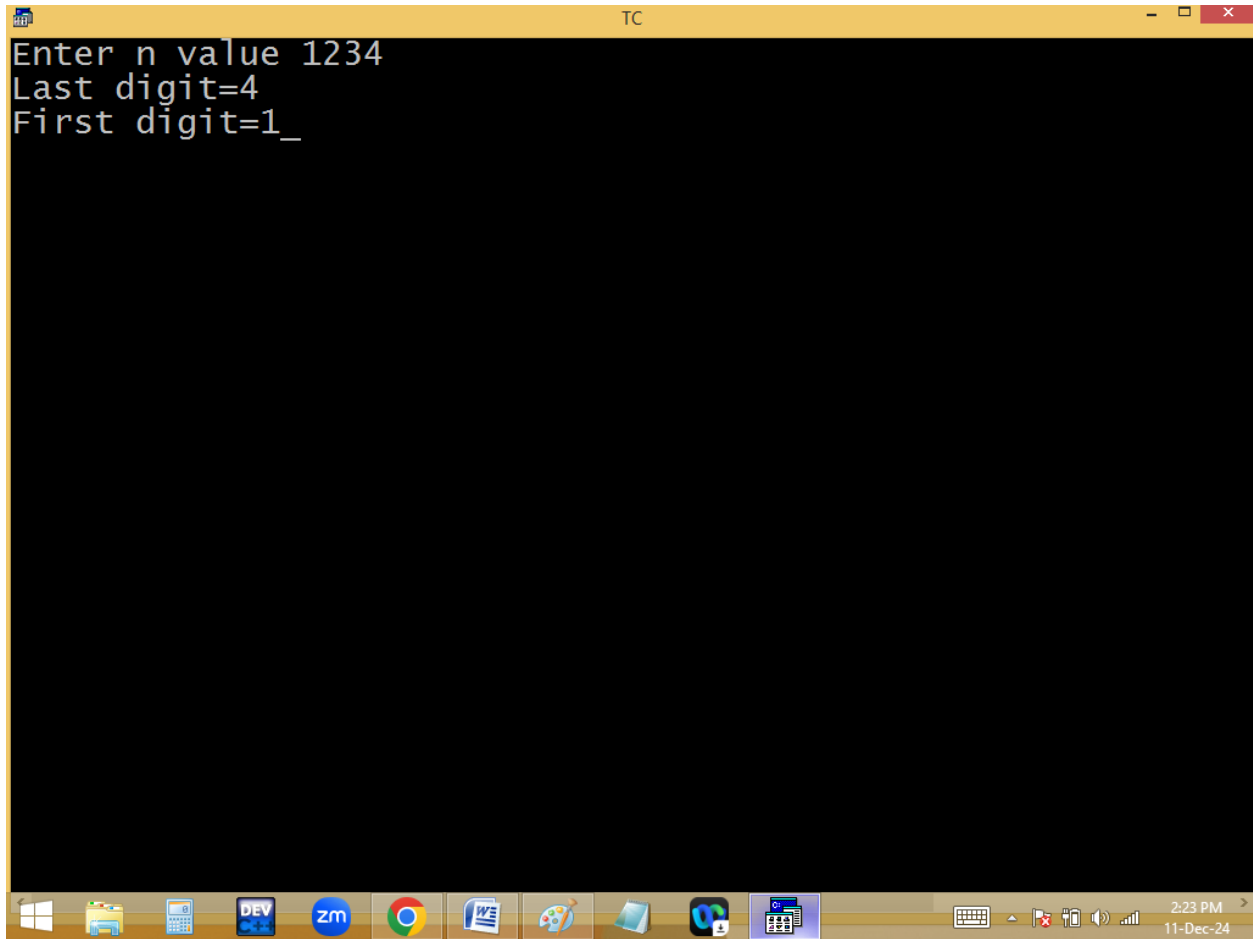


The image shows a screenshot of a Turbo C++ (TC) IDE window. The window has a yellow title bar with the text "TC" and standard window controls. Below the title bar is a menu bar with the following options: File, Edit, Run, Compile, Project, Options, and Debug. A status bar at the top of the editor area shows "Line 9 Col 16" and a list of keyboard shortcuts: Insert, Indent, Tab, Fill, Unindent, *, and E. The main editor area has a blue background and contains the following C code:

```
#include<stdio.h>
#include<conio.h>
void main()
{
long n;
clrscr();
printf("Enter n value "); scanf("%ld",&n);
printf("Last digit=%d\n",n%10);
while(n>9||n<-9)n=n/10;
printf("First digit=%d",n);
getch();
}
```

Below the editor area is a toolbar with function key shortcuts: F1-Help, F5-Zoom, F6-Switch, F7-Trace, F8-Step, F9-Make, and F10-Run. At the very bottom is the Windows taskbar, which includes icons for the Start menu, File Explorer, Calculator, DEV C++, Zoom, Google Chrome, Notepad, Paint, and the Turbo C++ application. The system clock in the bottom right corner shows "2:23 PM" and "11-Dec-24".

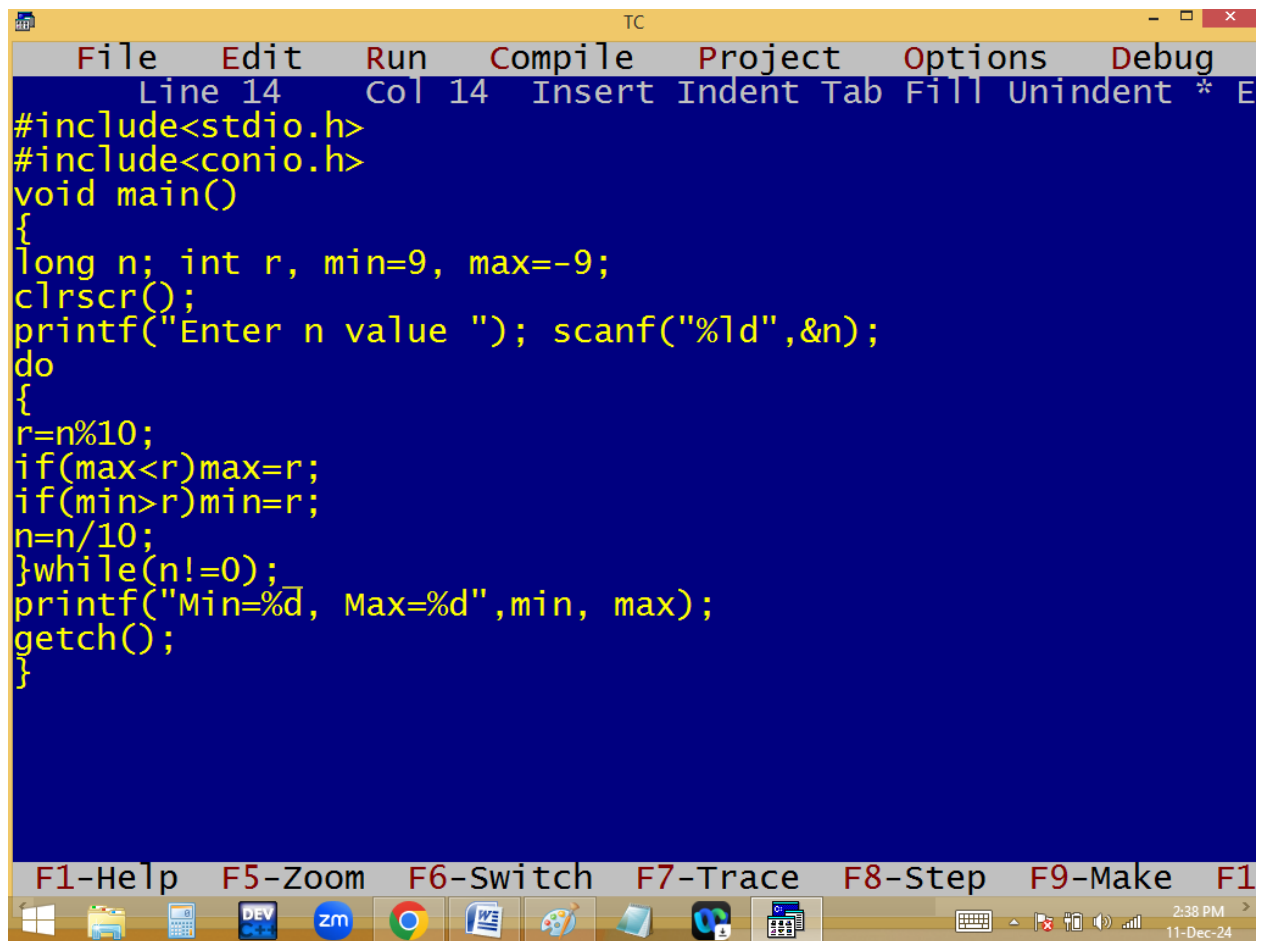




```
Enter n value 0
Last digit=0
First digit=0_
```

`p("Last digit=%d\n",n%10);` $\overline{n}$
`while( n>9 ) n/=10;` $1234 \% 10 = 4$
`p(n);`

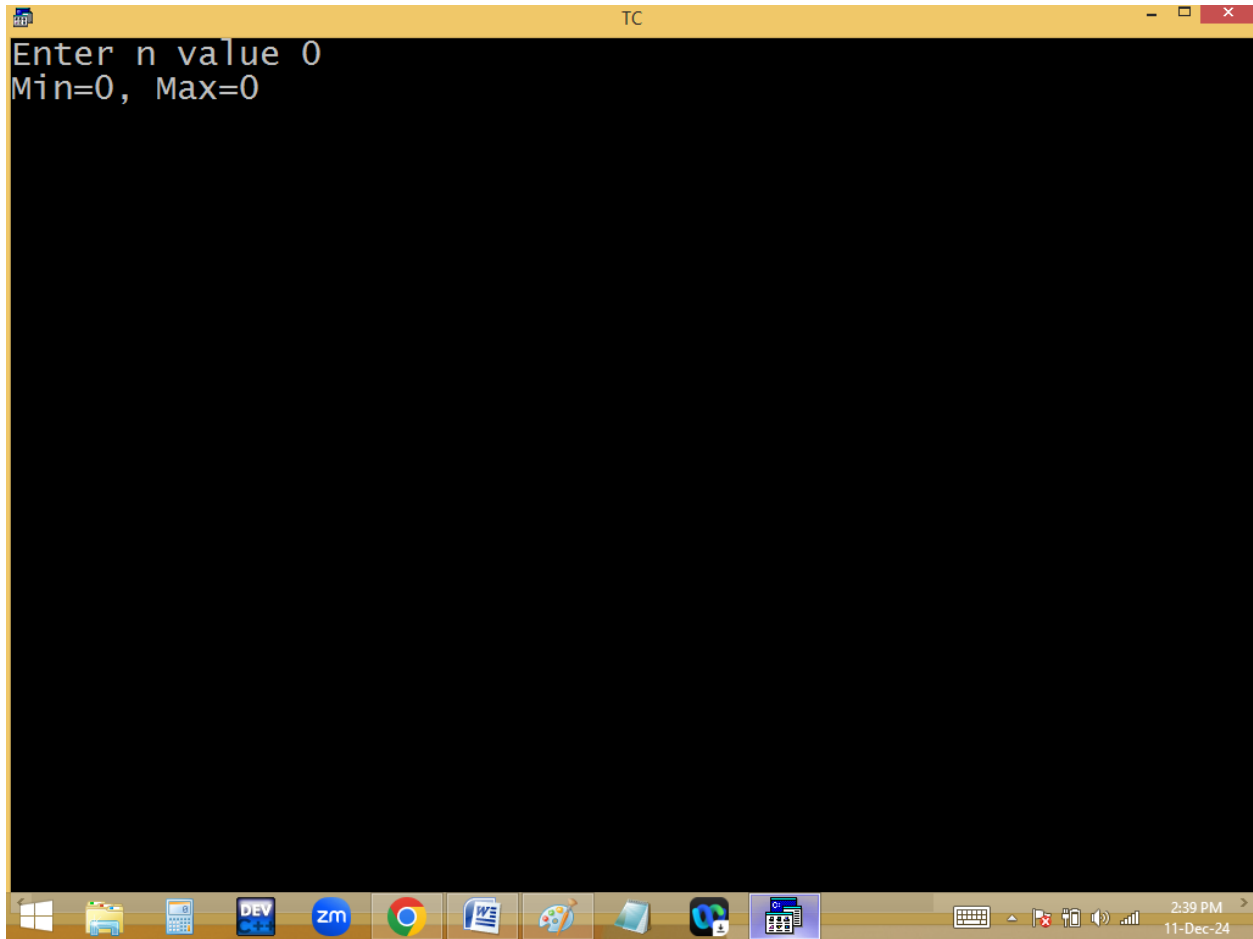
Finding max, min digits of given no:

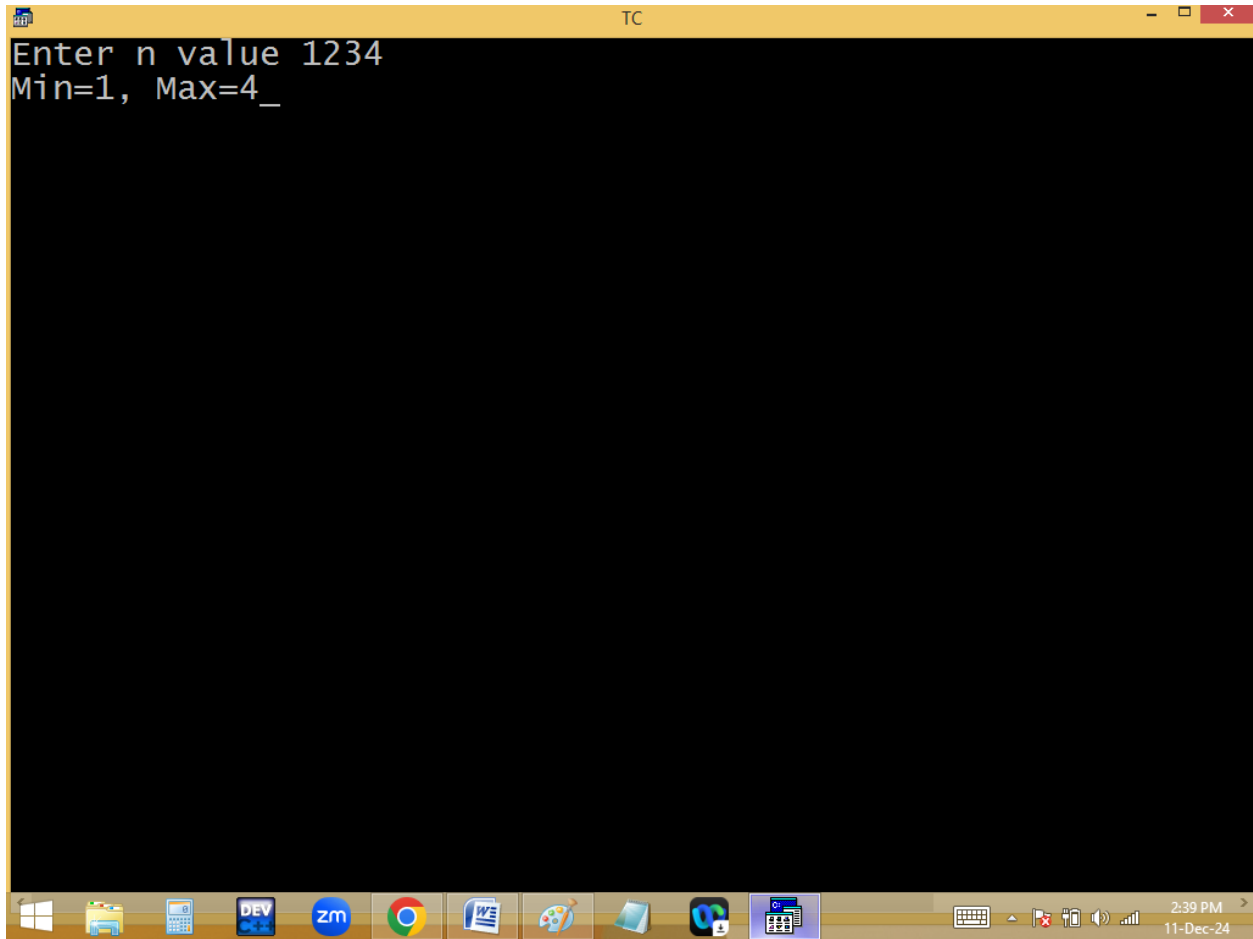


The image shows a screenshot of a Turbo C++ (TC) IDE window. The window has a yellow title bar with the text "TC" and standard window controls. Below the title bar is a menu bar with the following options: File, Edit, Run, Compile, Project, Options, and Debug. Under the "Run" menu, there is a submenu with the following options: Line 14, Col 14, Insert, Indent, Tab, Fill, Unindent, and *. Below the menu bar is a status bar with the text "Line 14 Col 14 Insert Indent Tab Fill Unindent *". The main area of the window is a blue background with yellow text, displaying the following C code:

```
#include<stdio.h>
#include<conio.h>
void main()
{
long n; int r, min=9, max=-9;
clrscr();
printf("Enter n value "); scanf("%ld",&n);
do
{
r=n%10;
if(max<r)max=r;
if(min>r)min=r;
n=n/10;
}while(n!=0);
printf("Min=%d, Max=%d",min, max);
getch();
}
```

At the bottom of the window, there is a toolbar with the following icons: F1-Help, F5-Zoom, F6-Switch, F7-Trace, F8-Step, F9-Make, and F1. Below the toolbar is a Windows taskbar with the following icons: Windows Start button, File Explorer, Calculator, DEV C++, Zoom, Google Chrome, Microsoft Word, Paint, and a folder icon. The taskbar also shows the system clock as 2:38 PM on 11-Dec-24.

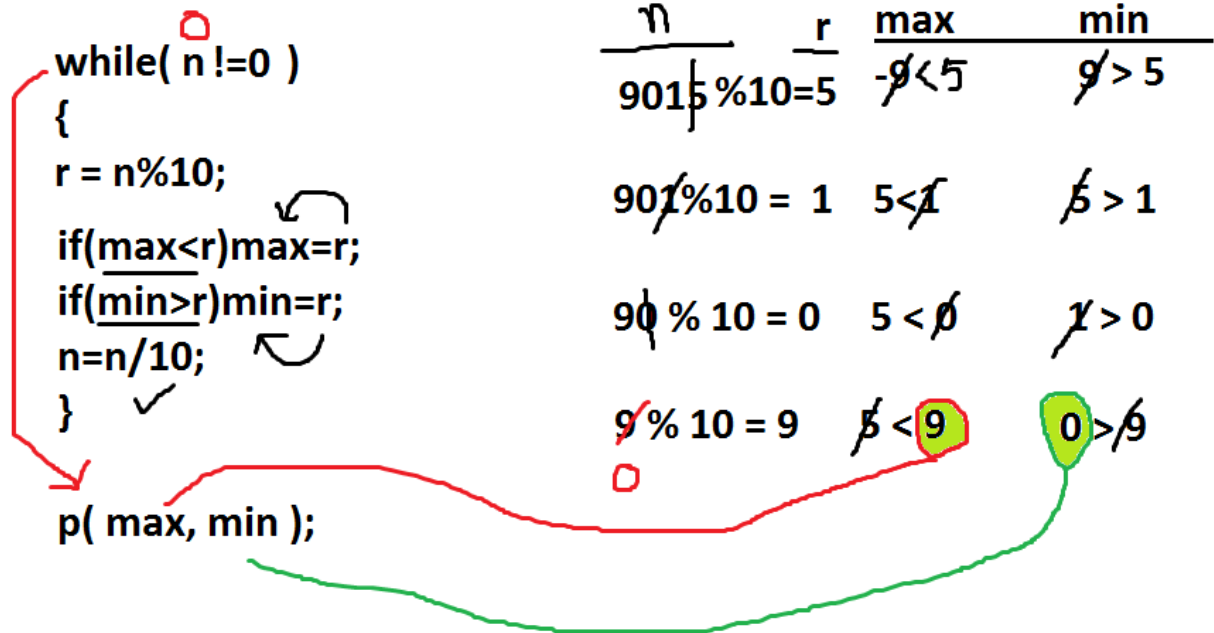





```

Enter n value -4567
Min=-7, Max=-4

```



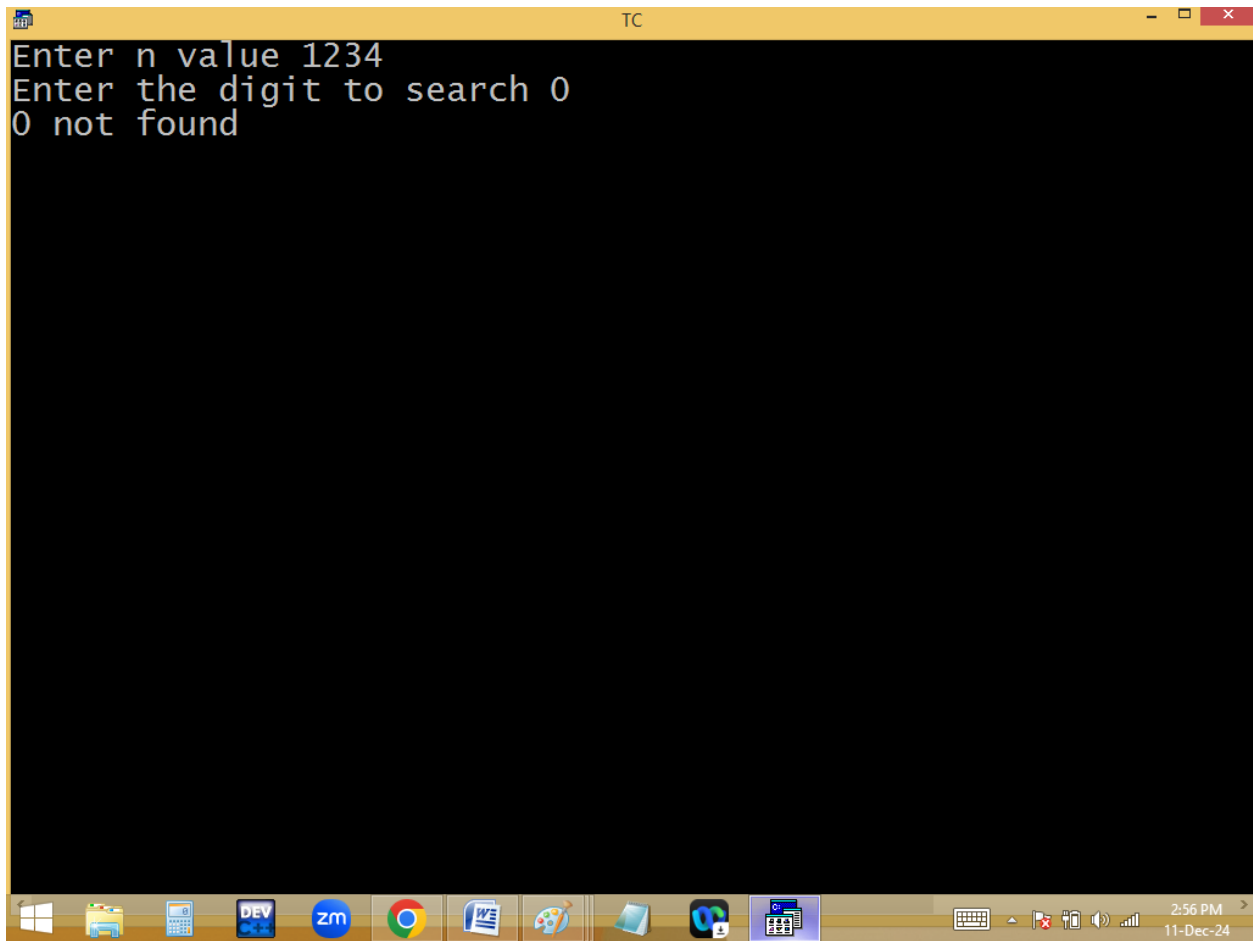
Finding the given digit position:

3015 → 0 position is 2

```
TC
File Edit Run Compile Project Options Debug
Line 5 Col 34 Insert Indent Tab Fill Unindent * E
#include<stdio.h>
#include<conio.h>
void main()
{
long n,m; int d,r,c=0,f=0;
clrscr();
printf("Enter n value "); scanf("%ld",&n);m=n;
printf("Enter the digit to search ");scanf("%d",&d);
do{c++;m=m/10;}while(m!=0); /* counting no of digits */
do
{
r=n%10;
if(r==d)printf("%d in %d position\n",d,c,f=1);
c--;
n=n/10;
}while(n!=0);
if(f==0)printf("%d not found",d);
getch();
}
```

F1-Help F5-Zoom F6-Switch F7-Trace F8-Step F9-Make F10-Run

2:56 PM
11-Dec-24



```

Enter n value 102030
Enter the digit to search 0
0 in 6 position
0 in 4 position
0 in 2 position

```

while(m) {c++;m=m/10;}

$m = n$

<u>d</u>	<u>c:=0</u>
0	4 1
	3 2
	2 3
	4

3015 % 10 = 5
 301 % 10 = 1
 30 % 10 = 0

while(n!=0)

```

{
  r=n%10;
  if(r==d) printf("%d in %d position\n",d, c);
  c--; ✓
  n=n/10; ✓
}

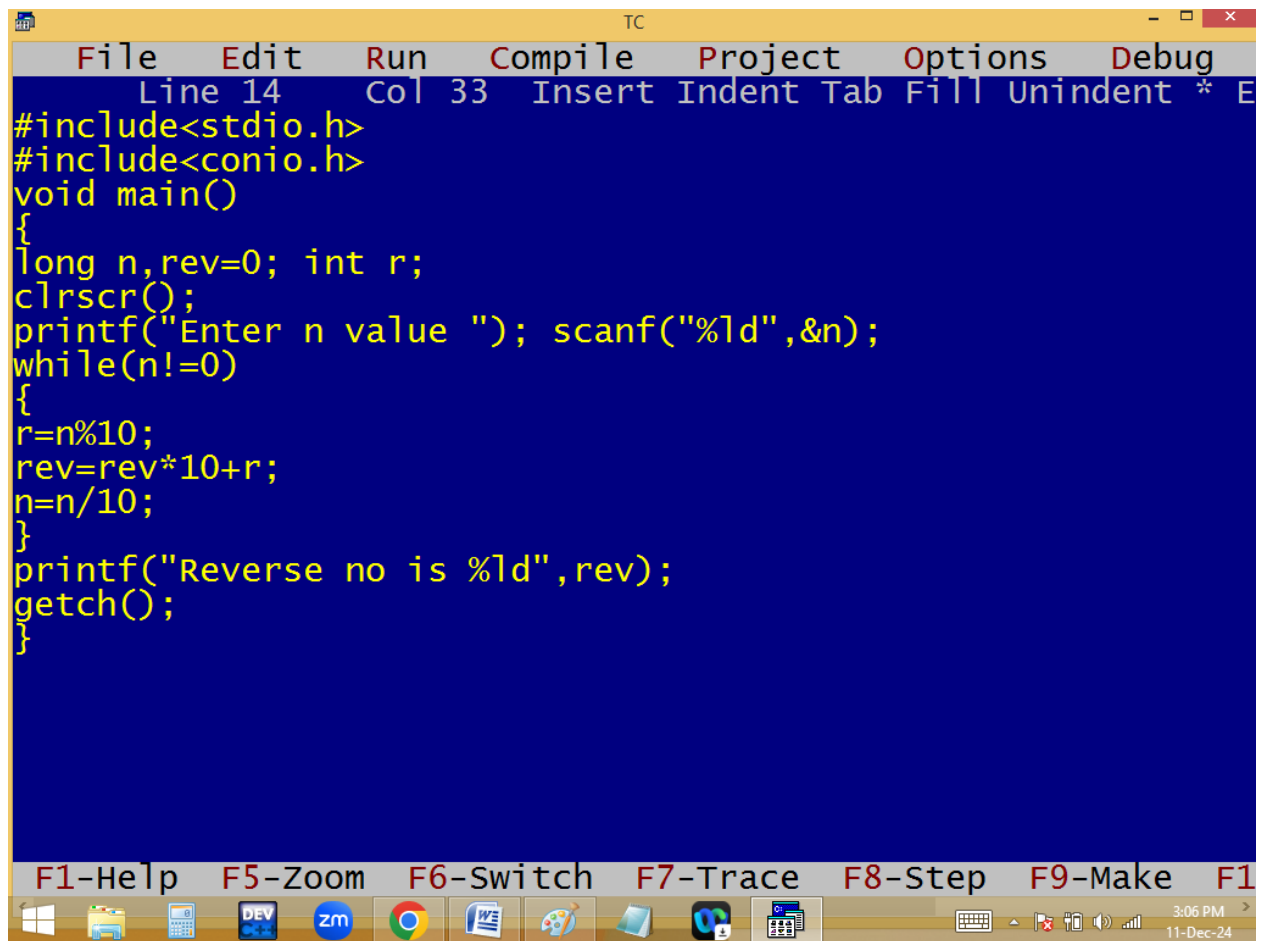
```

0 2

3015
 4
 2

Reverse no:

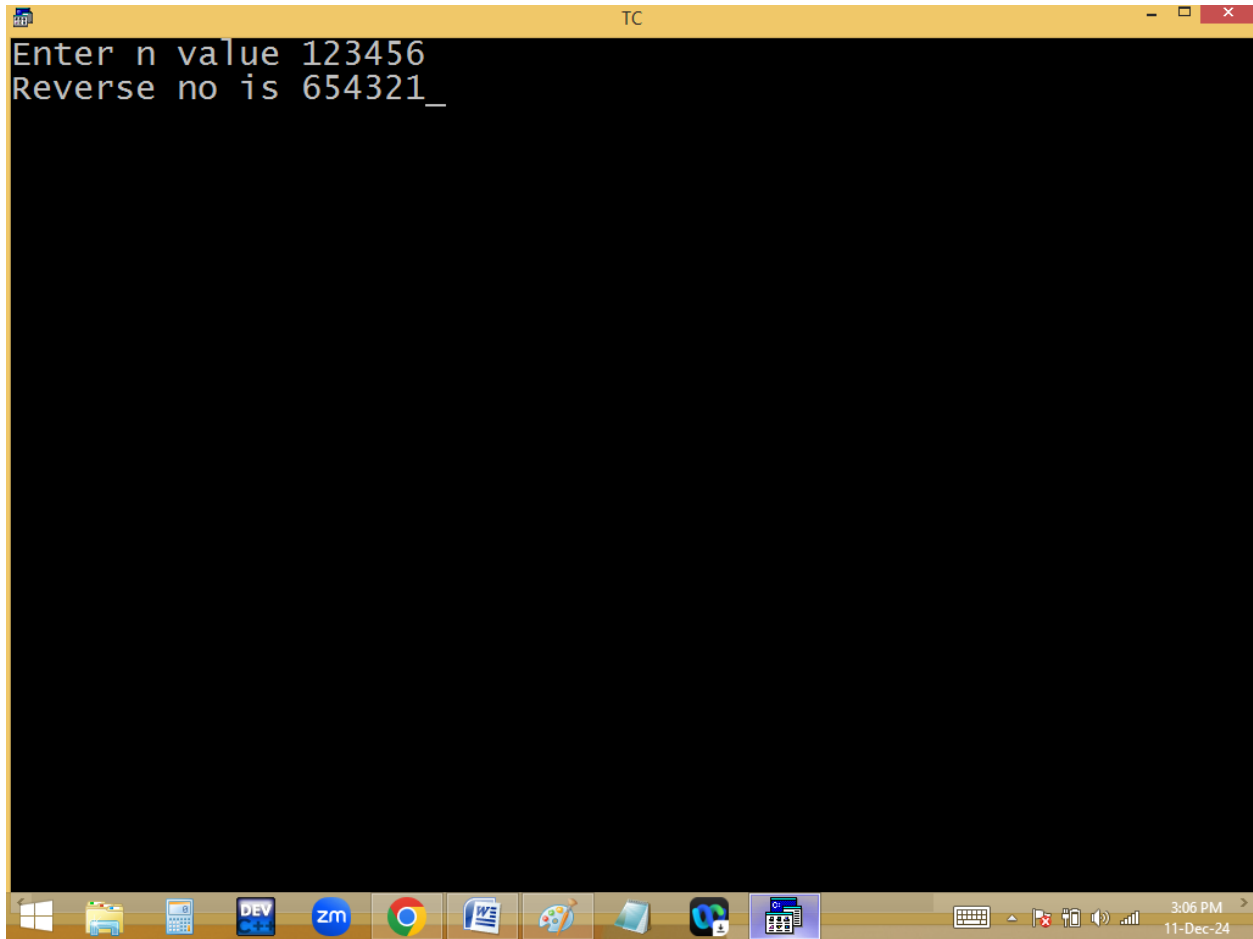
$$\begin{array}{r} \underline{11} \\ 139 \end{array} \begin{array}{l} \rightarrow 9 \times 10 = 90 \\ \rightarrow \frac{+3}{93 \times 10 = 930} \\ \rightarrow \frac{+1}{931} \end{array}$$

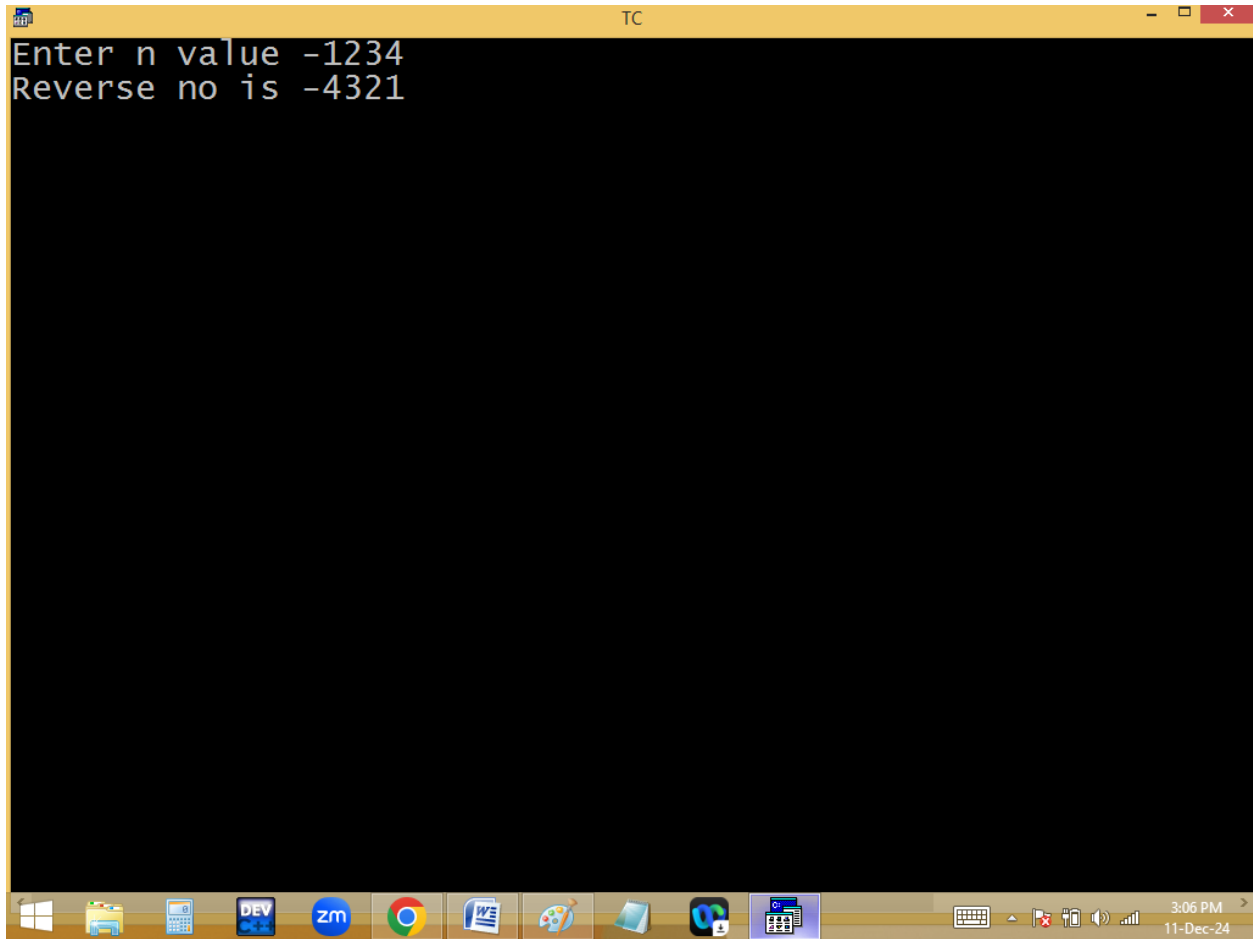


The image shows a screenshot of the Turbo C++ (TC) integrated development environment. The window title is "TC". The menu bar includes "File", "Edit", "Run", "Compile", "Project", "Options", and "Debug". Below the menu bar, the status bar shows "Line 14", "Col 33", and a list of editing options: "Insert", "Indent", "Tab", "Fill", "Unindent", and a "Format" icon. The main editing area has a dark blue background with yellow text. The code is a C program to reverse a number using a while loop. The code is as follows:

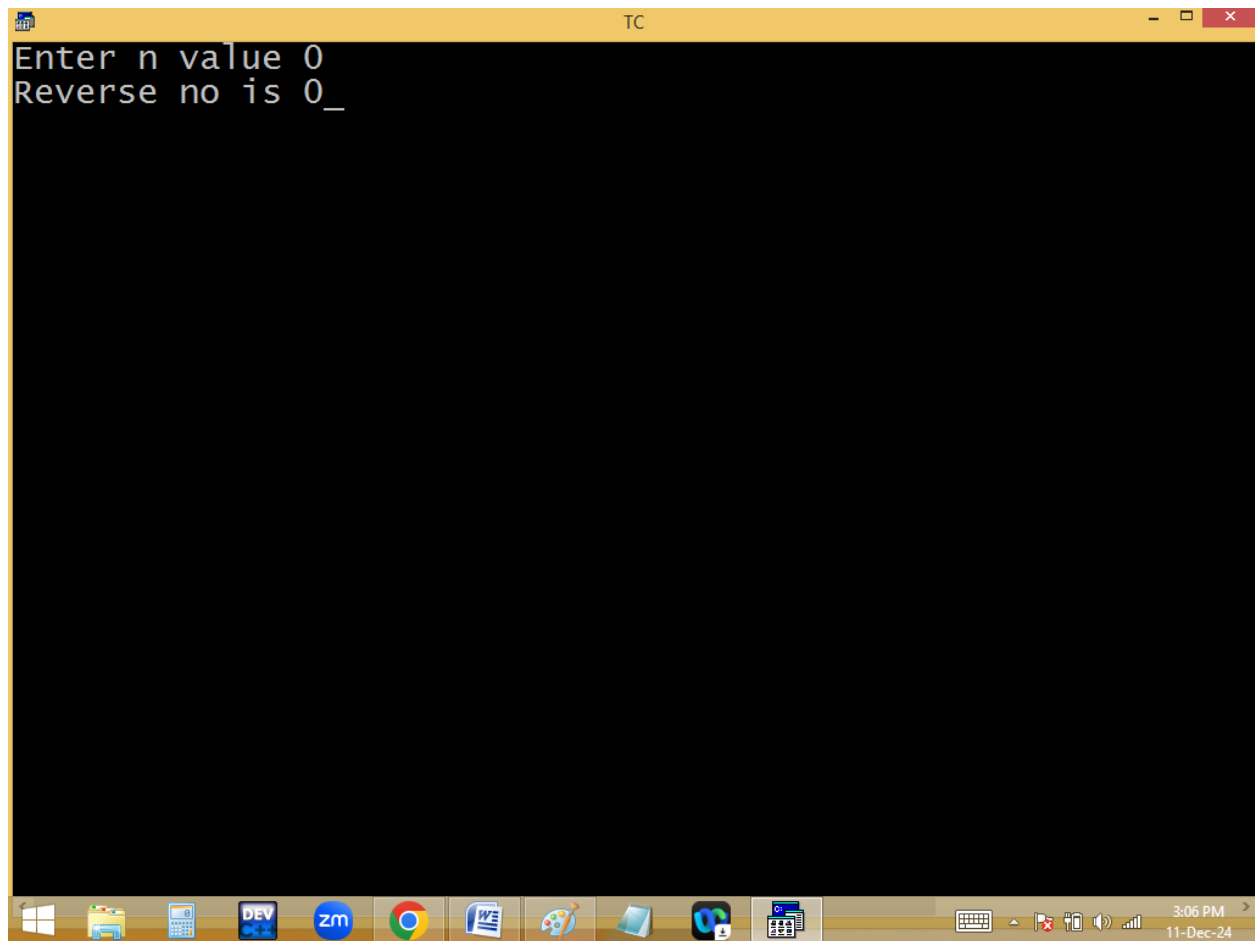
```
#include<stdio.h>
#include<conio.h>
void main()
{
    long n,rev=0; int r;
    clrscr();
    printf("Enter n value "); scanf("%ld",&n);
    while(n!=0)
    {
        r=n%10;
        rev=rev*10+r;
        n=n/10;
    }
    printf("Reverse no is %ld",rev);
    getch();
}
```

At the bottom of the window, there is a toolbar with function key shortcuts: "F1-Help", "F5-Zoom", "F6-Switch", "F7-Trace", "F8-Step", "F9-Make", and "F10-Run". The Windows taskbar is visible at the very bottom, showing icons for the Start menu, File Explorer, Calculator, DEV C++, Zoom, Google Chrome, Word, Paint, and other applications. The system clock in the bottom right corner shows "3:06 PM" and "11-Dec-24".



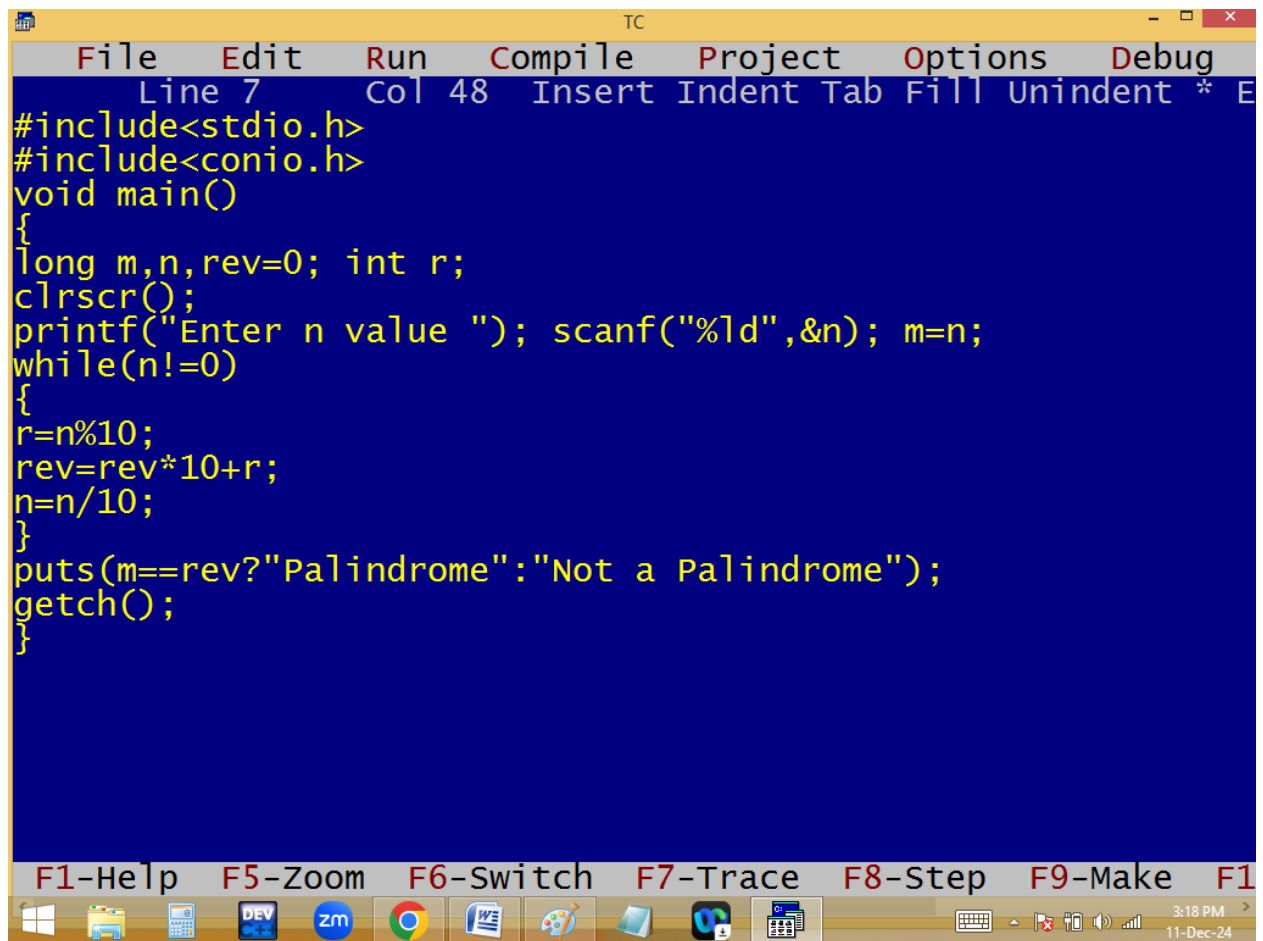


```
TC
Enter n value 0
Reverse no is 0_
```



Finding palindrome or not?

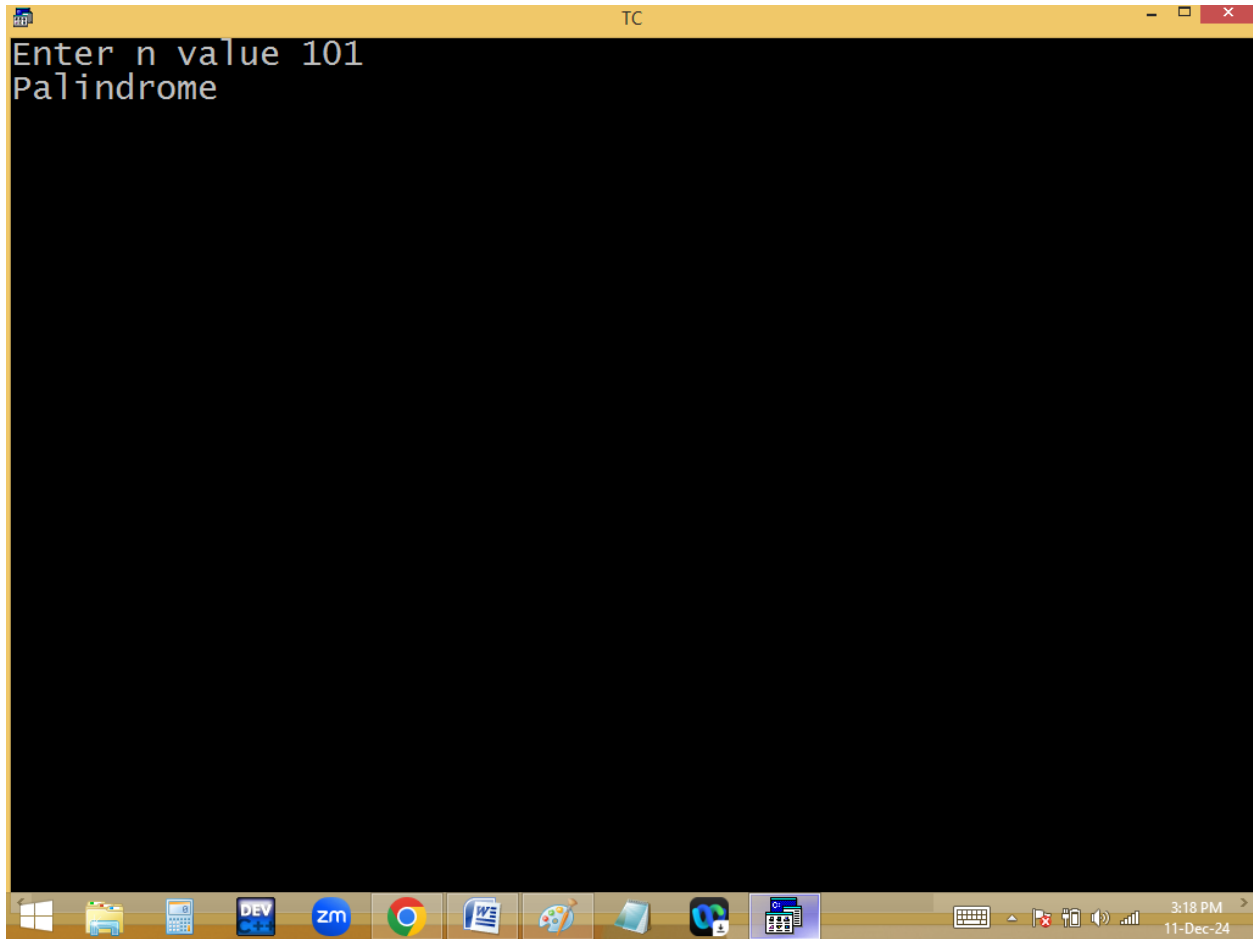
101 reverse is 101

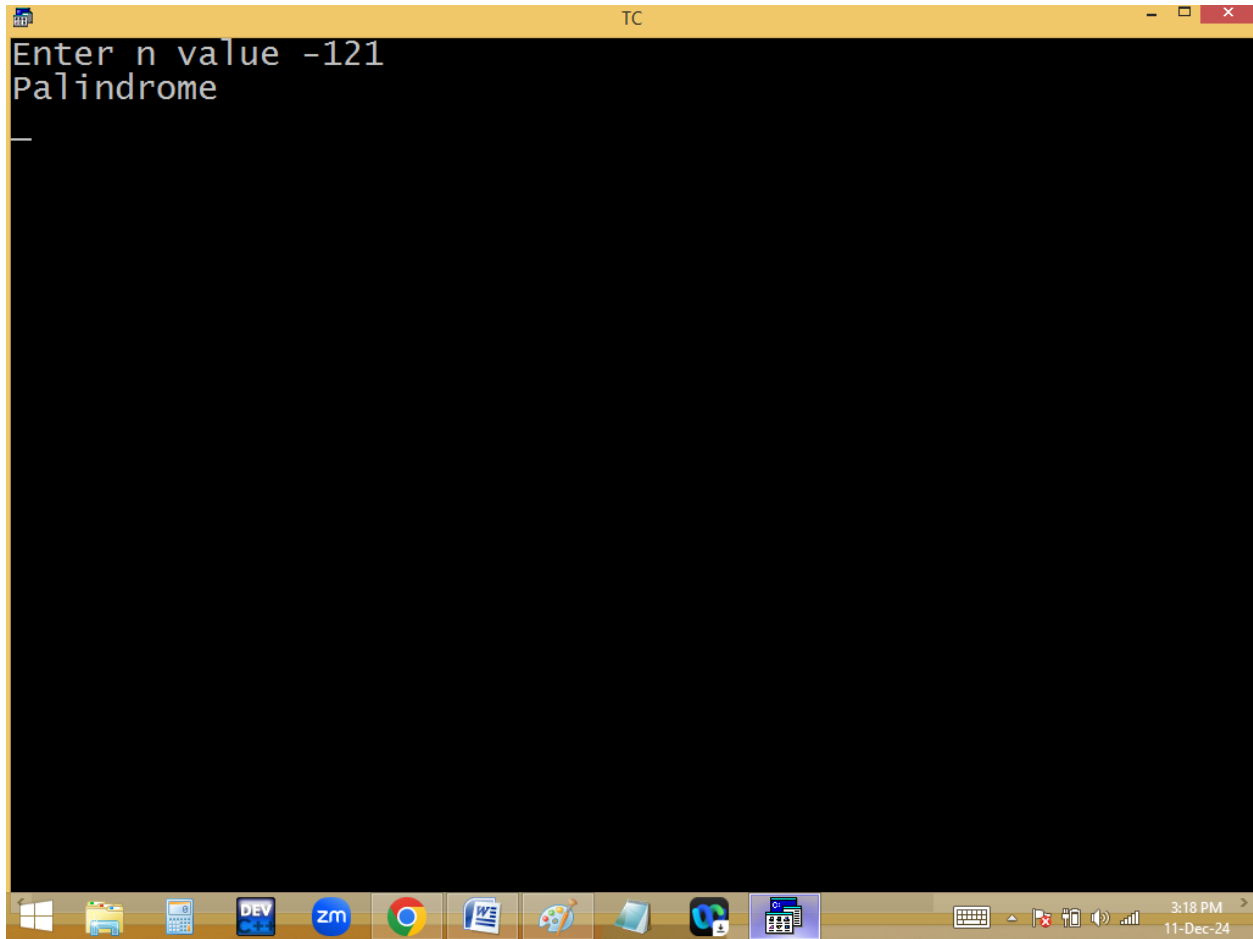


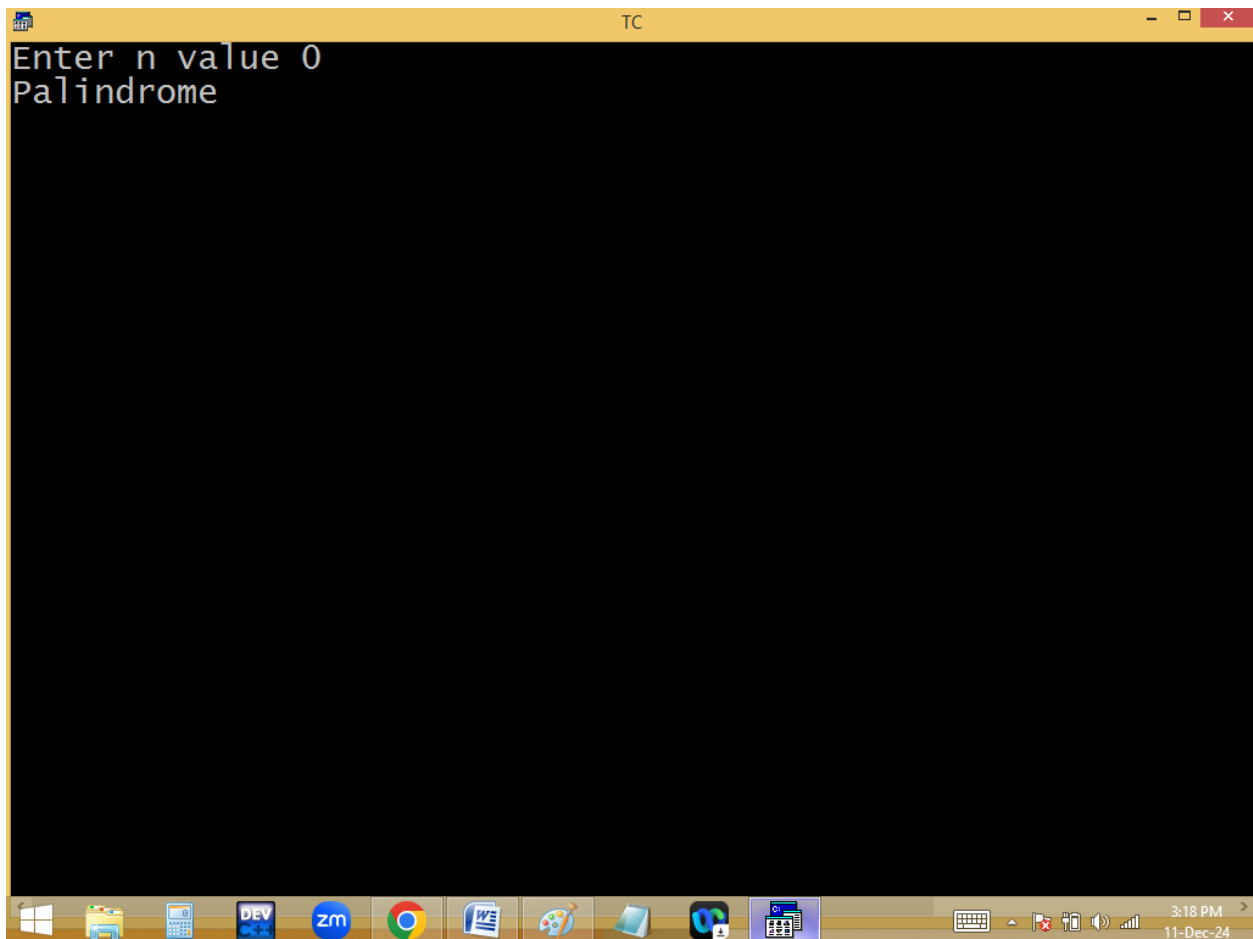
The image shows a screenshot of the Turbo C++ (TC) IDE. The window title is "TC". The menu bar includes "File", "Edit", "Run", "Compile", "Project", "Options", and "Debug". The status bar at the top indicates "Line 7", "Col 48", and lists editing options: "Insert", "Indent", "Tab", "Fill", "Unindent", and a "E" (likely for End). The main editing area has a blue background and contains the following C code:

```
#include<stdio.h>
#include<conio.h>
void main()
{
    long m,n,rev=0; int r;
    clrscr();
    printf("Enter n value "); scanf("%ld",&n); m=n;
    while(n!=0)
    {
        r=n%10;
        rev=rev*10+r;
        n=n/10;
    }
    puts(m==rev?"Palindrome":"Not a Palindrome");
    getch();
}
```

Below the code editor, there is a toolbar with function key shortcuts: "F1-Help", "F5-Zoom", "F6-Switch", "F7-Trace", "F8-Step", "F9-Make", and "F10-Run". The Windows taskbar is visible at the bottom, showing icons for the Start menu, File Explorer, Calculator, DEV C++, Zoom, Google Chrome, Word, Paint, and other applications. The system clock in the bottom right corner shows "3:18 PM" and "11-Dec-24".







$$\begin{array}{r} n \\ 139 \\ 139 \% 10 = 9 \\ 13 \% 10 = 3 \\ 1 \% 10 = 1 \\ 0 \end{array}$$

```

while( n!=0 )
{
    r = n%10; ✓
    rev = rev * 10 + r; ✓
    n=n/10; ✓ ✓
}

p( rev );

```

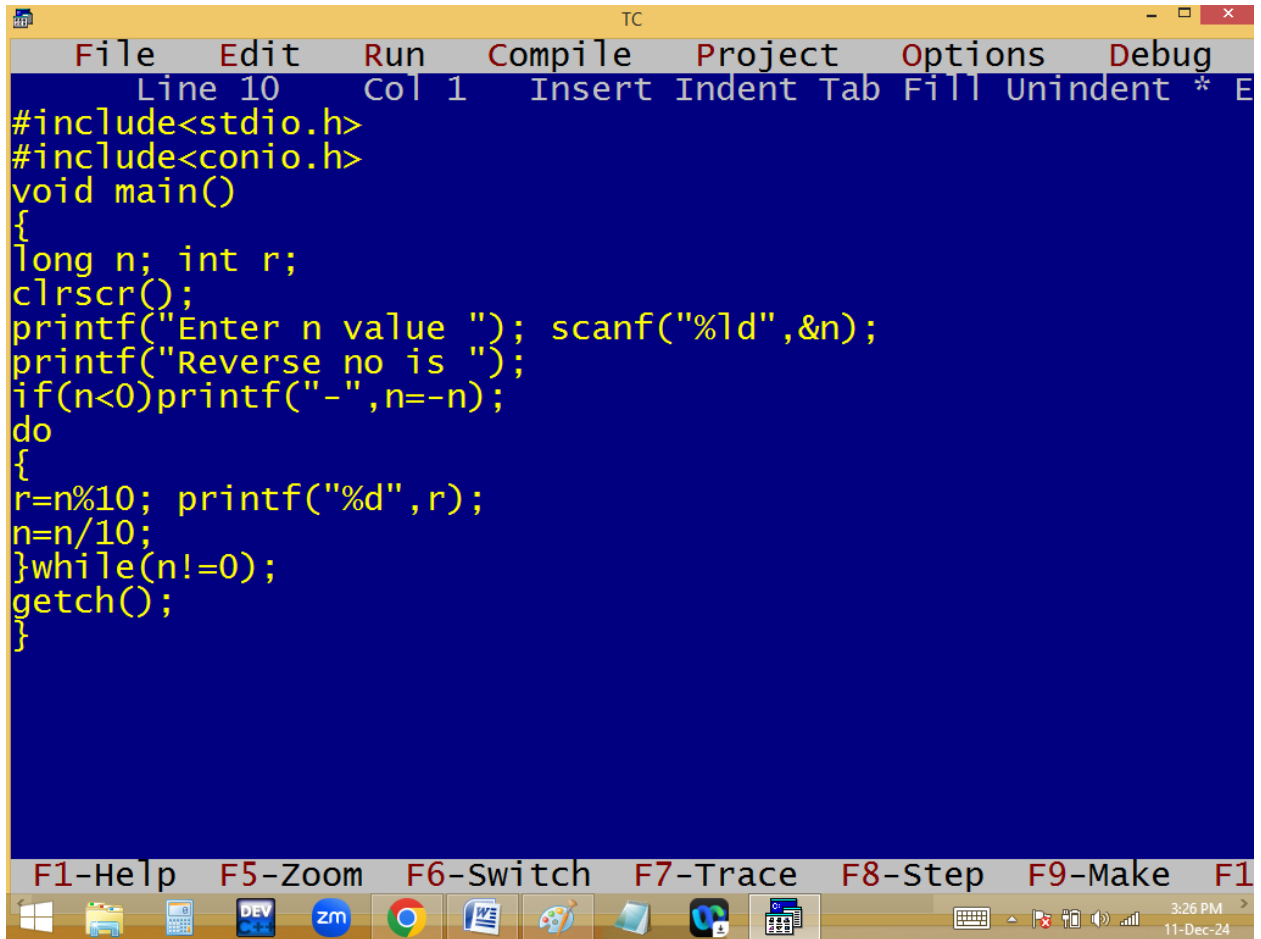
$$\begin{array}{r} rev \\ 0 \times 10 + 9 = 9 \\ 9 \times 10 + 3 = 93 \\ 93 \times 10 + 1 = 931 \end{array}$$

$$\begin{array}{r} f \leftarrow n \\ m = \frac{n}{10} \\ \frac{101}{10} = 10 \\ 10 \end{array}$$

$$\begin{array}{r} rev \\ 0 \times 10 + 1 = 1 \\ 1 \times 10 + 0 = 10 \\ 10 \times 10 + 1 = 101 \end{array}$$

if(m==rev)p(pal);
 else p(not);

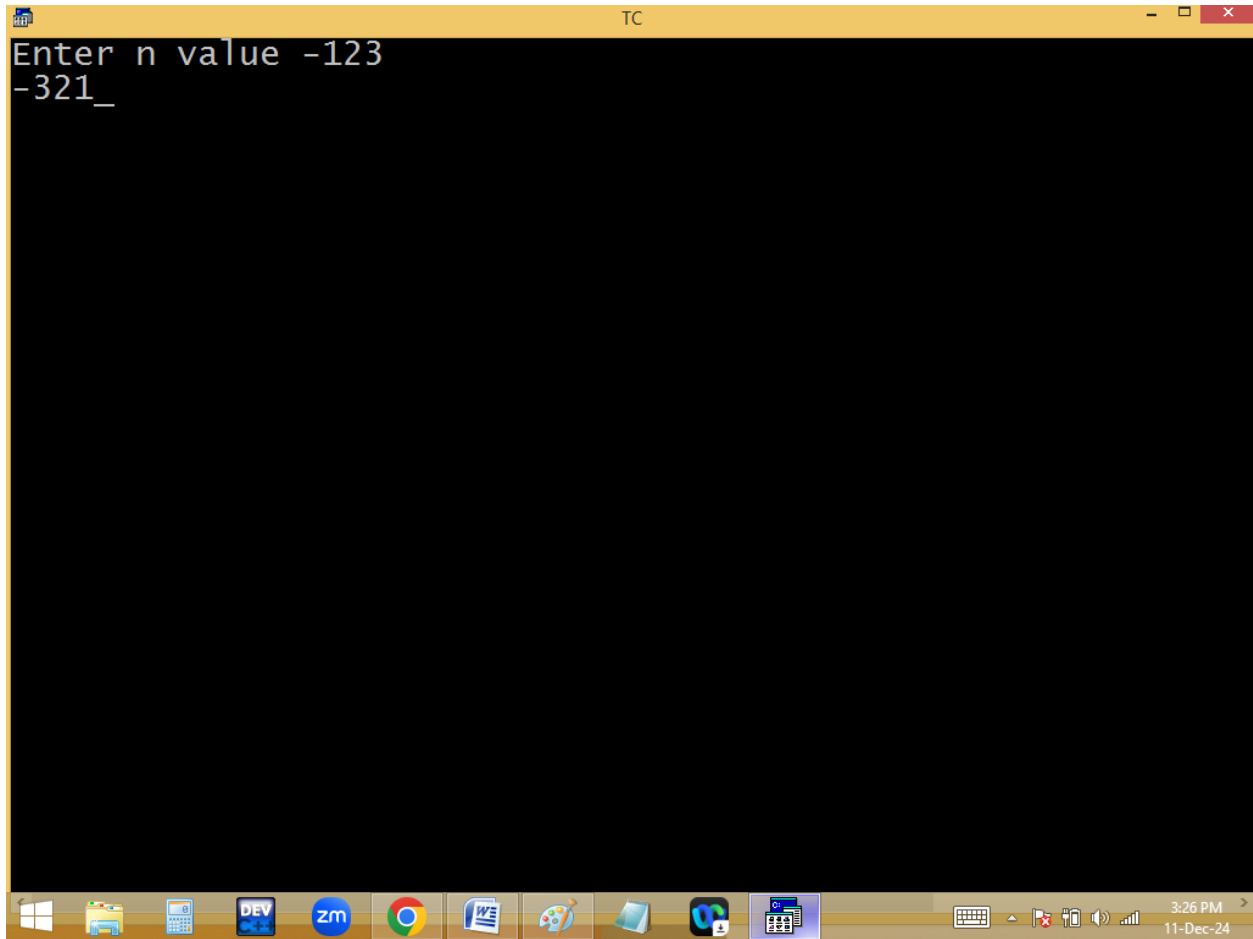
printing 100 as 001:

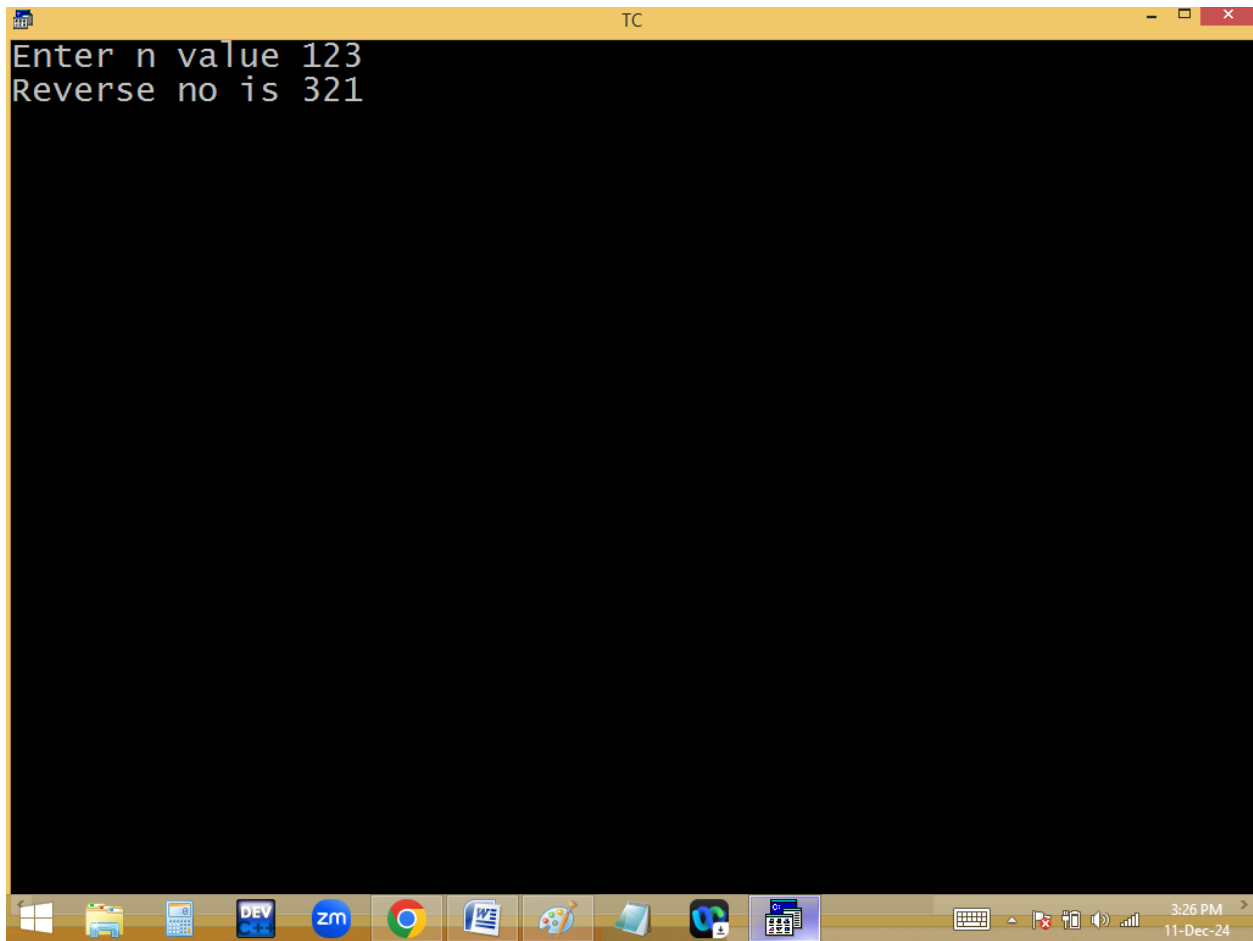


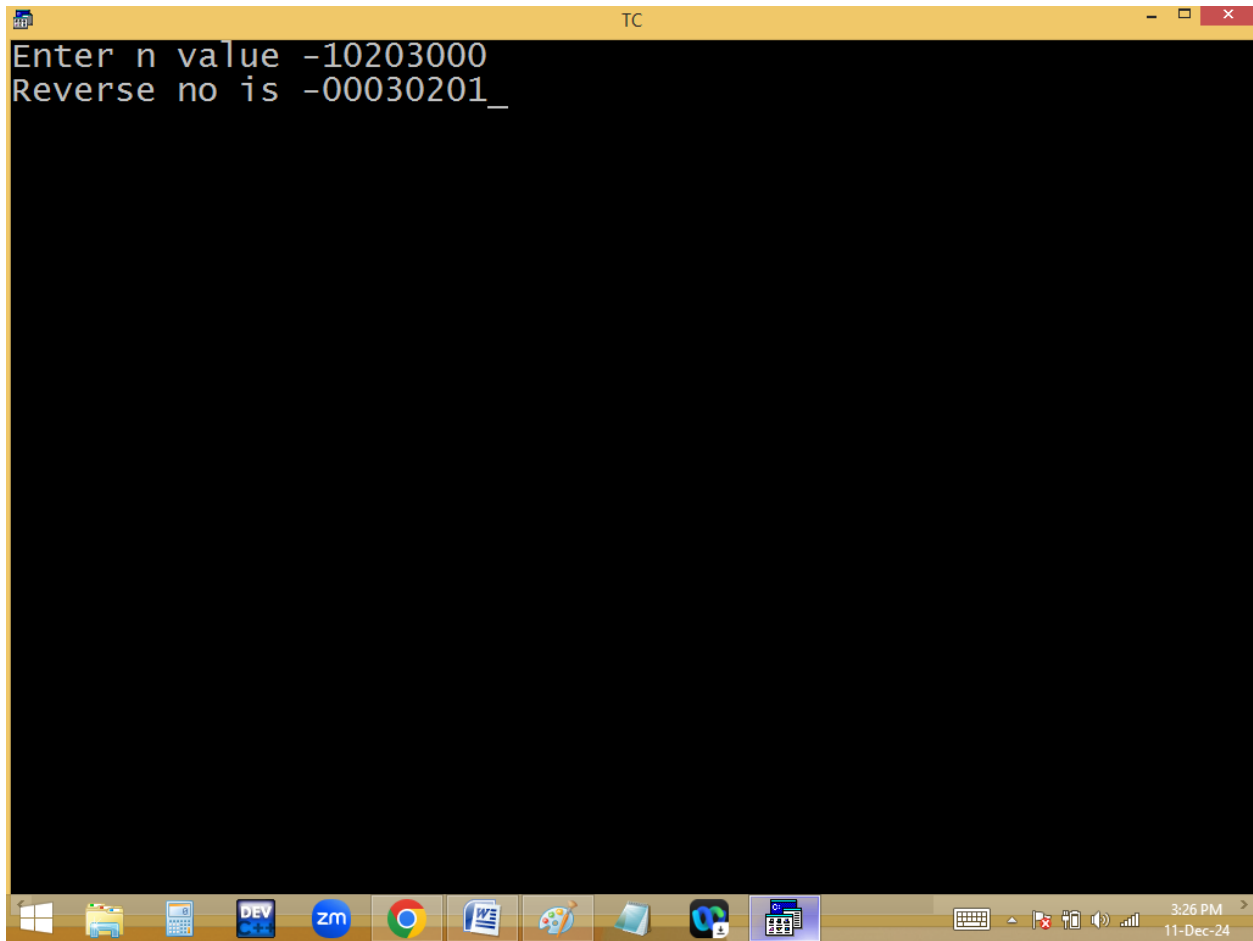
```
TC
File Edit Run Compile Project Options Debug
Line 10 Col 1 Insert Indent Tab Fill Unindent * E
#include<stdio.h>
#include<conio.h>
void main()
{
long n; int r;
clrscr();
printf("Enter n value "); scanf("%ld",&n);
printf("Reverse no is ");
if(n<0)printf("-",n=-n);
do
{
r=n%10; printf("%d",r);
n=n/10;
}while(n!=0);
getch();
}
```

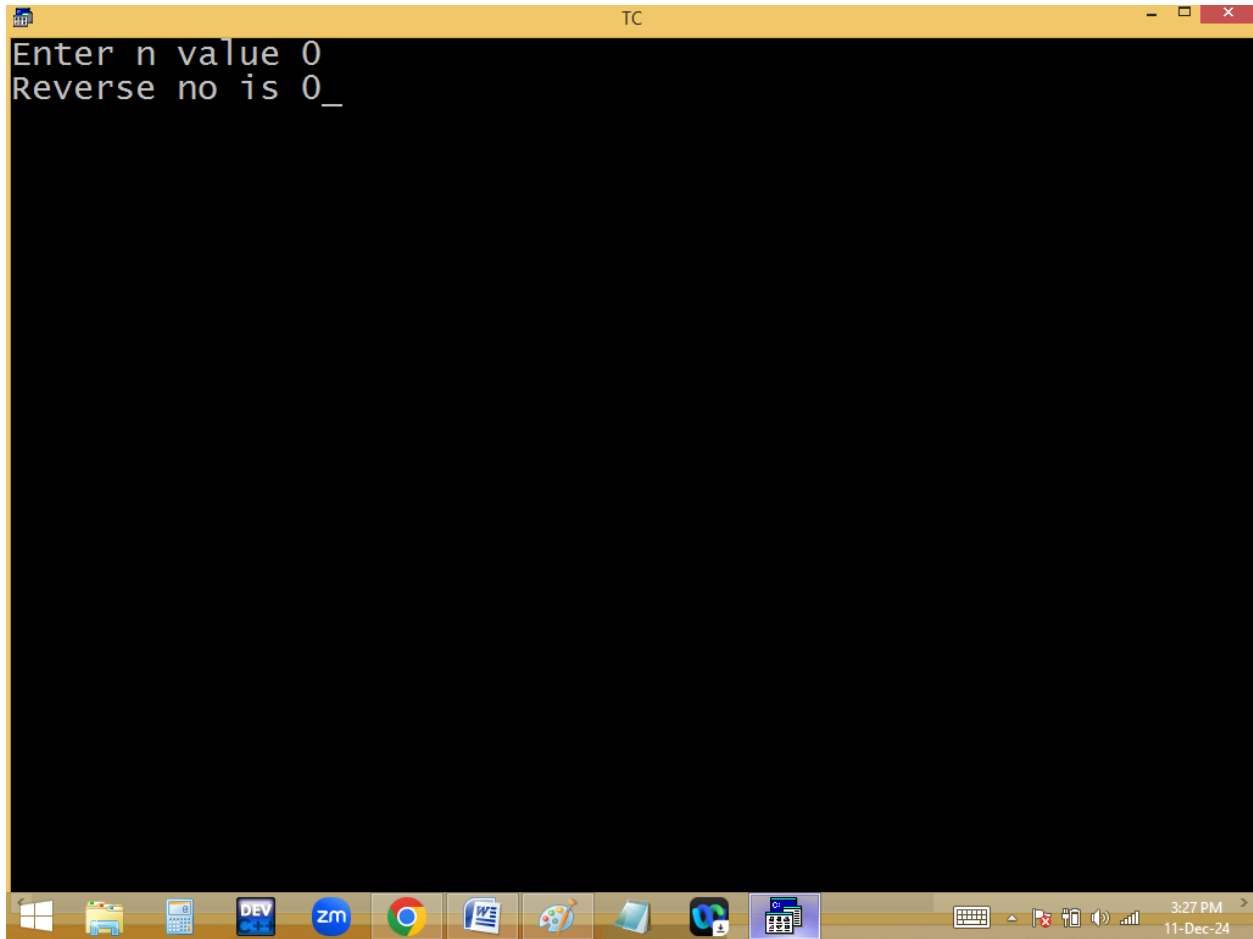
F1-Help F5-Zoom F6-Switch F7-Trace F8-Step F9-Make F10-Run

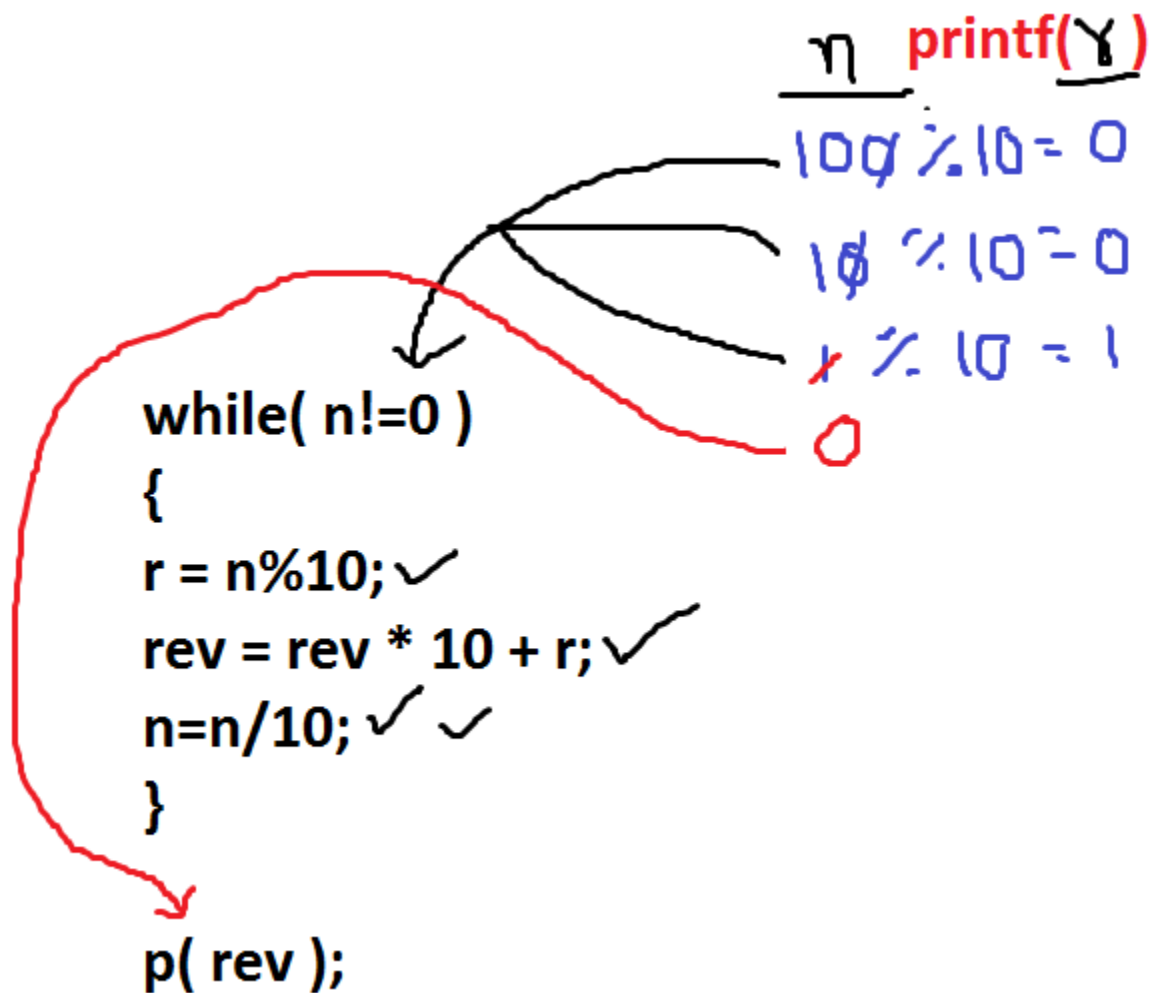
3:26 PM 11-Dec-24





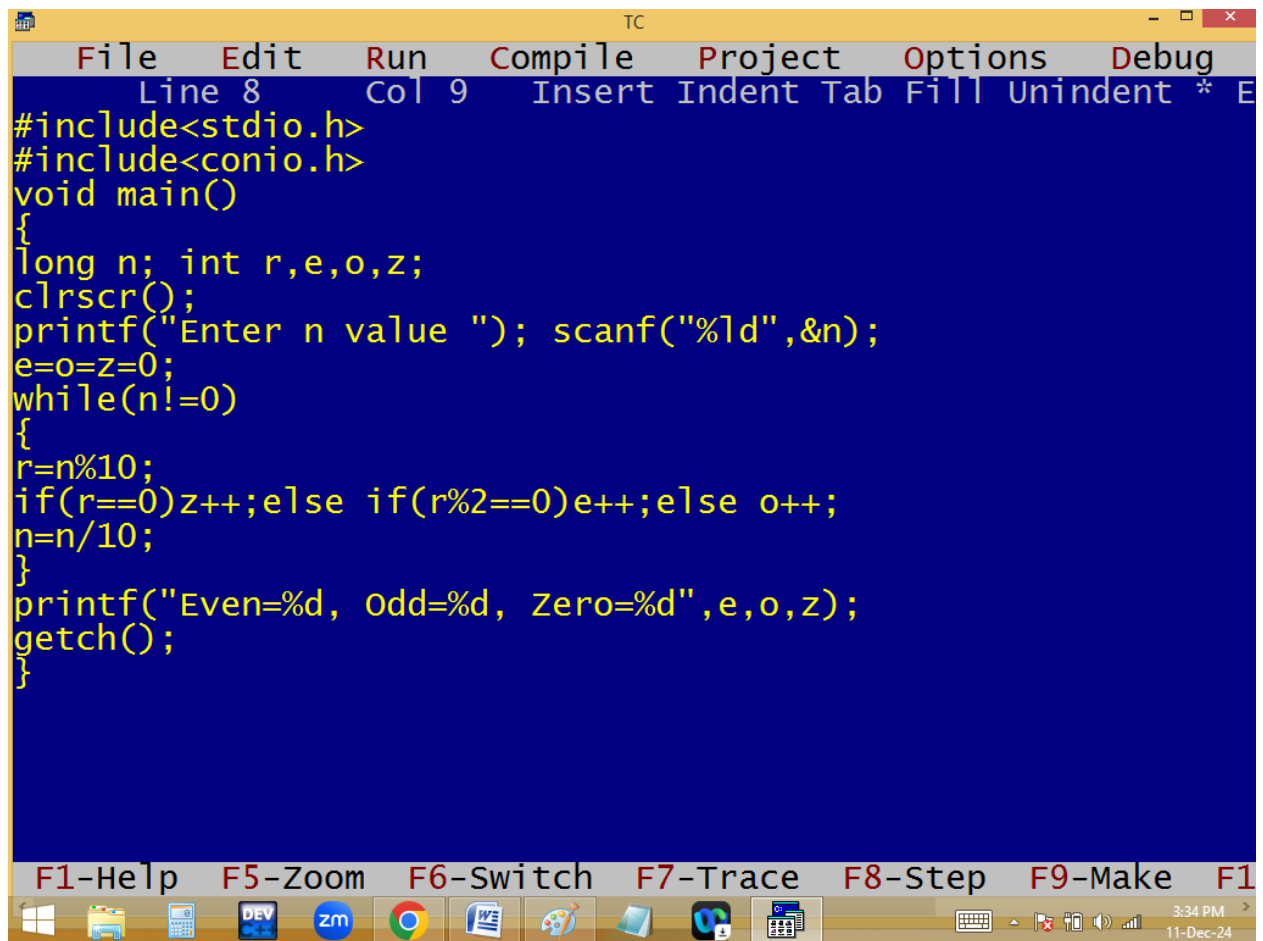






Finding no of even, odd, zero digits in given no

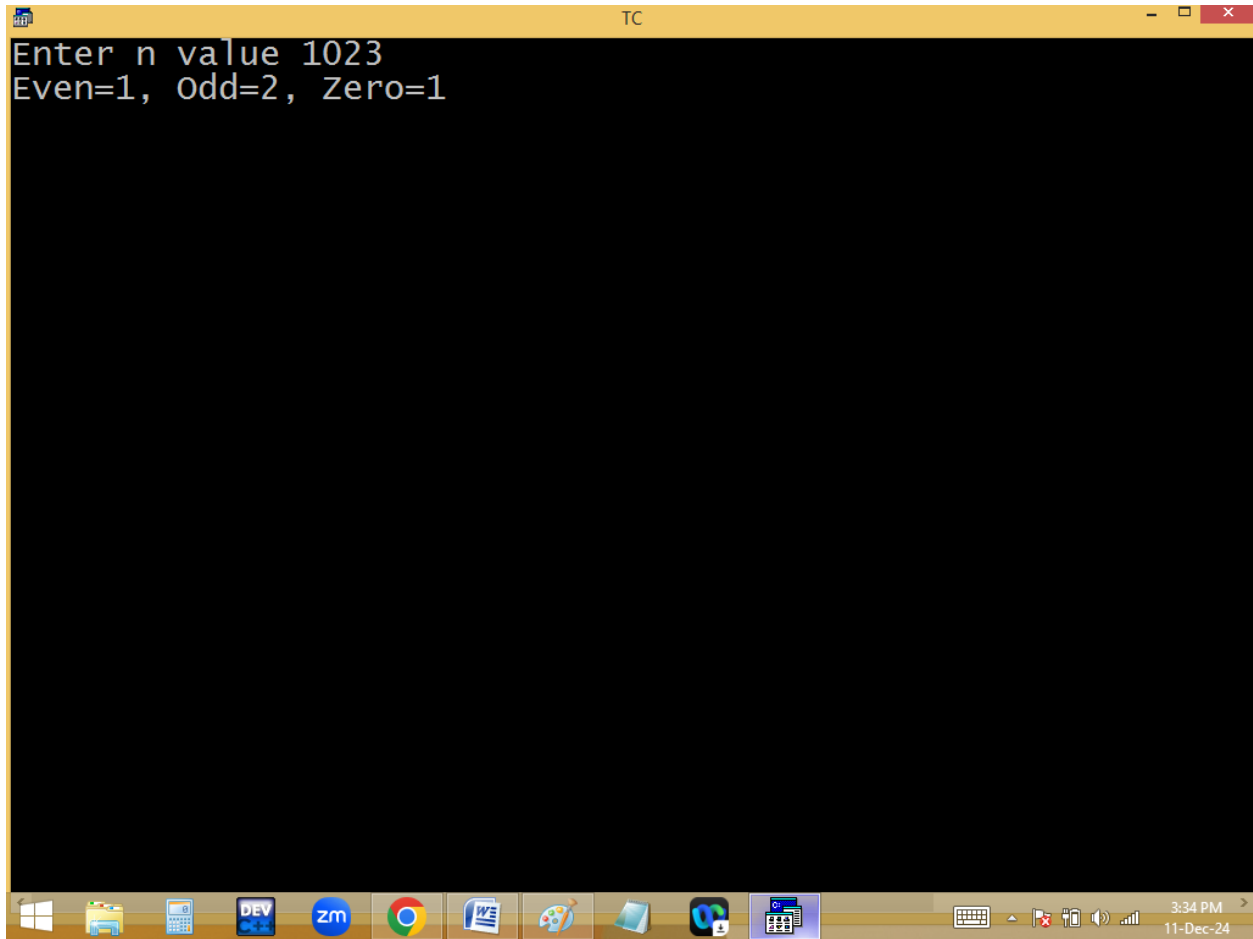
Eg: 1023 → 1 even, 2 odd, 1 zero

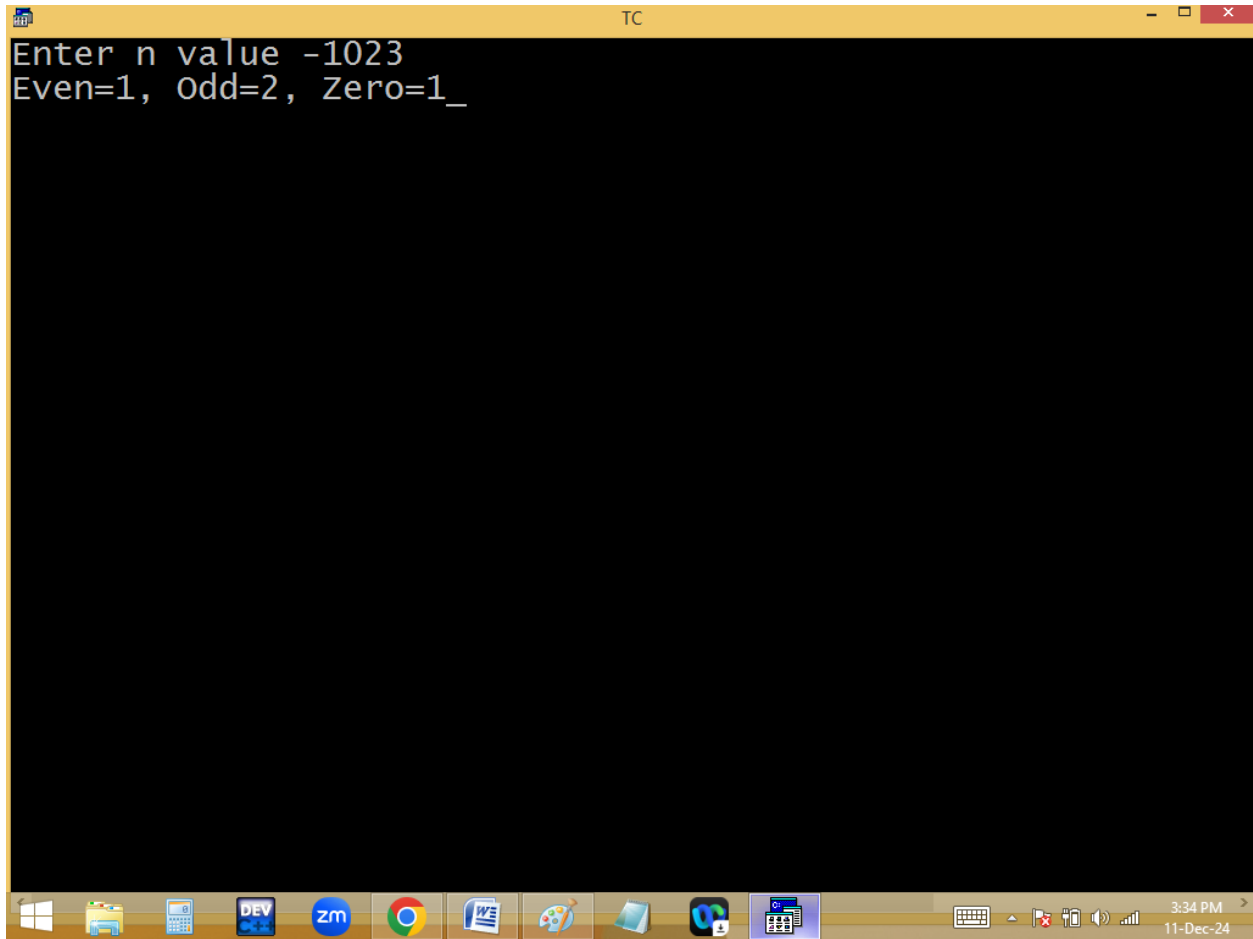


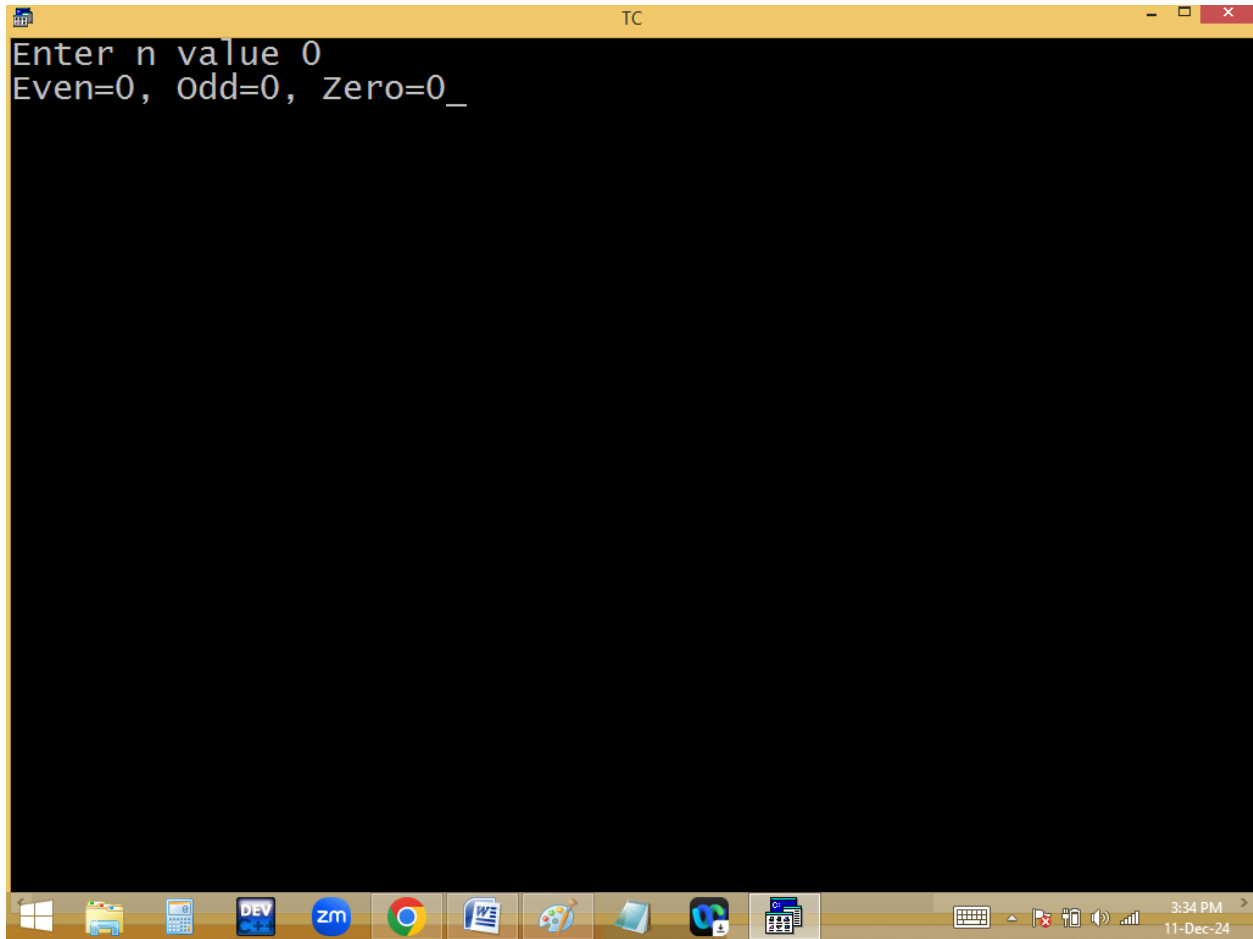
The image shows a screenshot of a Turbo C++ (TC) IDE window. The window has a yellow title bar with the text "TC" and standard window controls. Below the title bar is a menu bar with the following options: File, Edit, Run, Compile, Project, Options, and Debug. A status bar at the top of the editor area shows "Line 8", "Col 9", and "Insert Indent Tab Fill Unindent * E". The main editor area has a dark blue background with yellow text. The code is as follows:

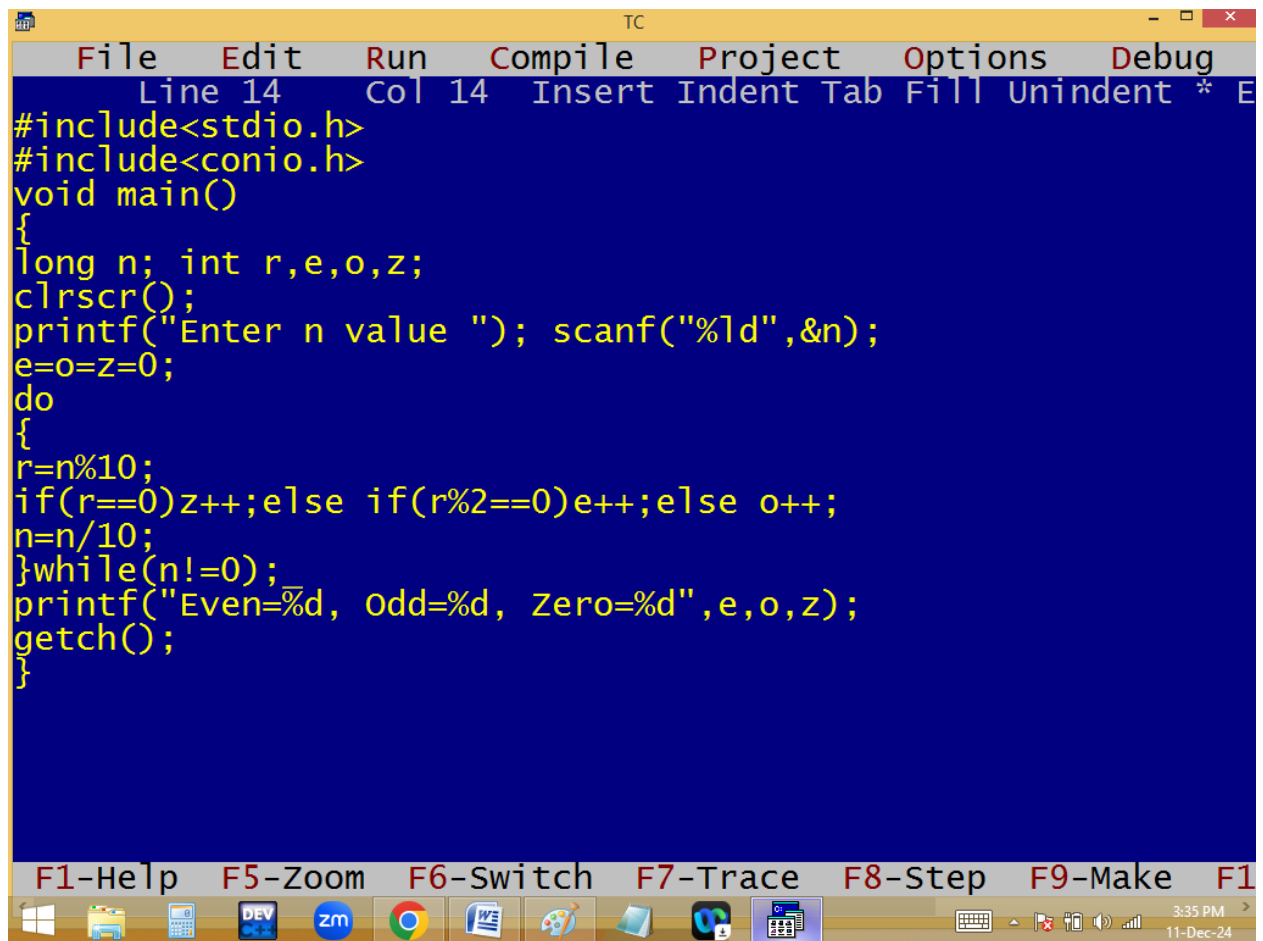
```
#include<stdio.h>
#include<conio.h>
void main()
{
long n; int r,e,o,z;
clrscr();
printf("Enter n value "); scanf("%ld",&n);
e=o=z=0;
while(n!=0)
{
r=n%10;
if(r==0)z++;else if(r%2==0)e++;else o++;
n=n/10;
}
printf("Even=%d, Odd=%d, Zero=%d",e,o,z);
getch();
}
```

Below the editor area is a toolbar with function key shortcuts: F1-Help, F5-Zoom, F6-Switch, F7-Trace, F8-Step, F9-Make, and F10-Run. At the very bottom is the Windows taskbar, which includes icons for the Start button, File Explorer, Calculator, DEV C++, Zoom, Google Chrome, Notepad, Paint, and several other applications. The system clock in the bottom right corner shows "3:34 PM" and "11-Dec-24".





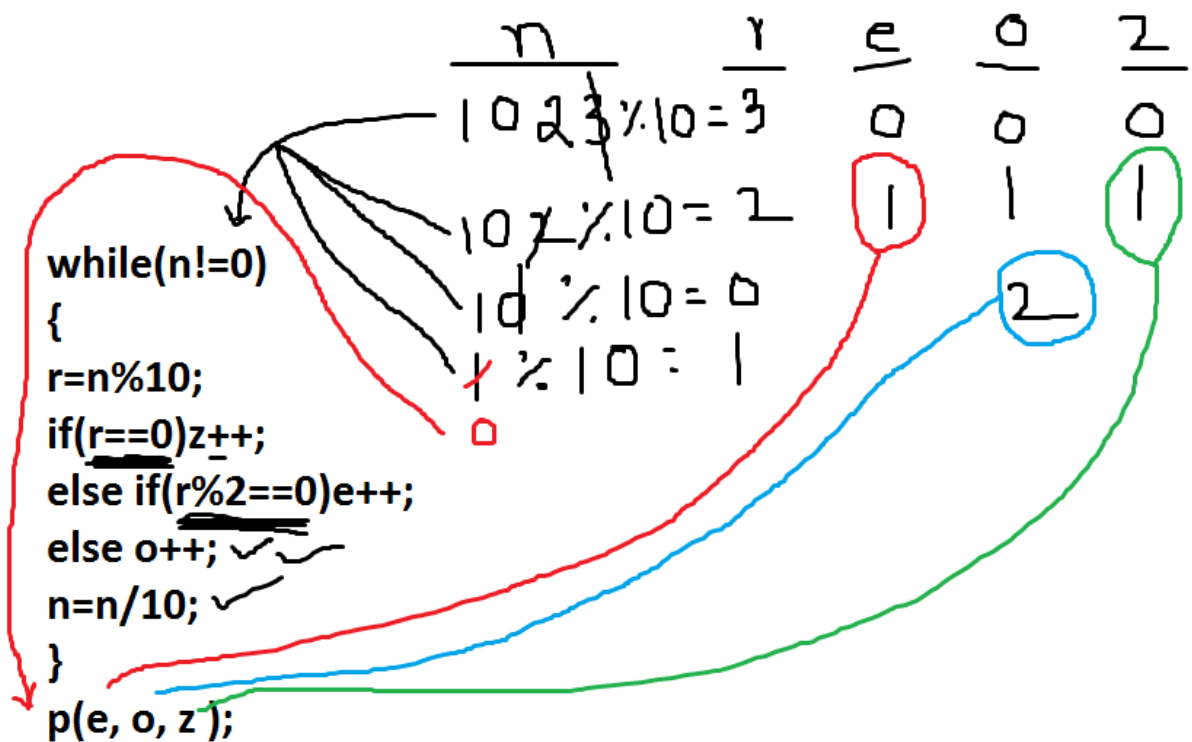
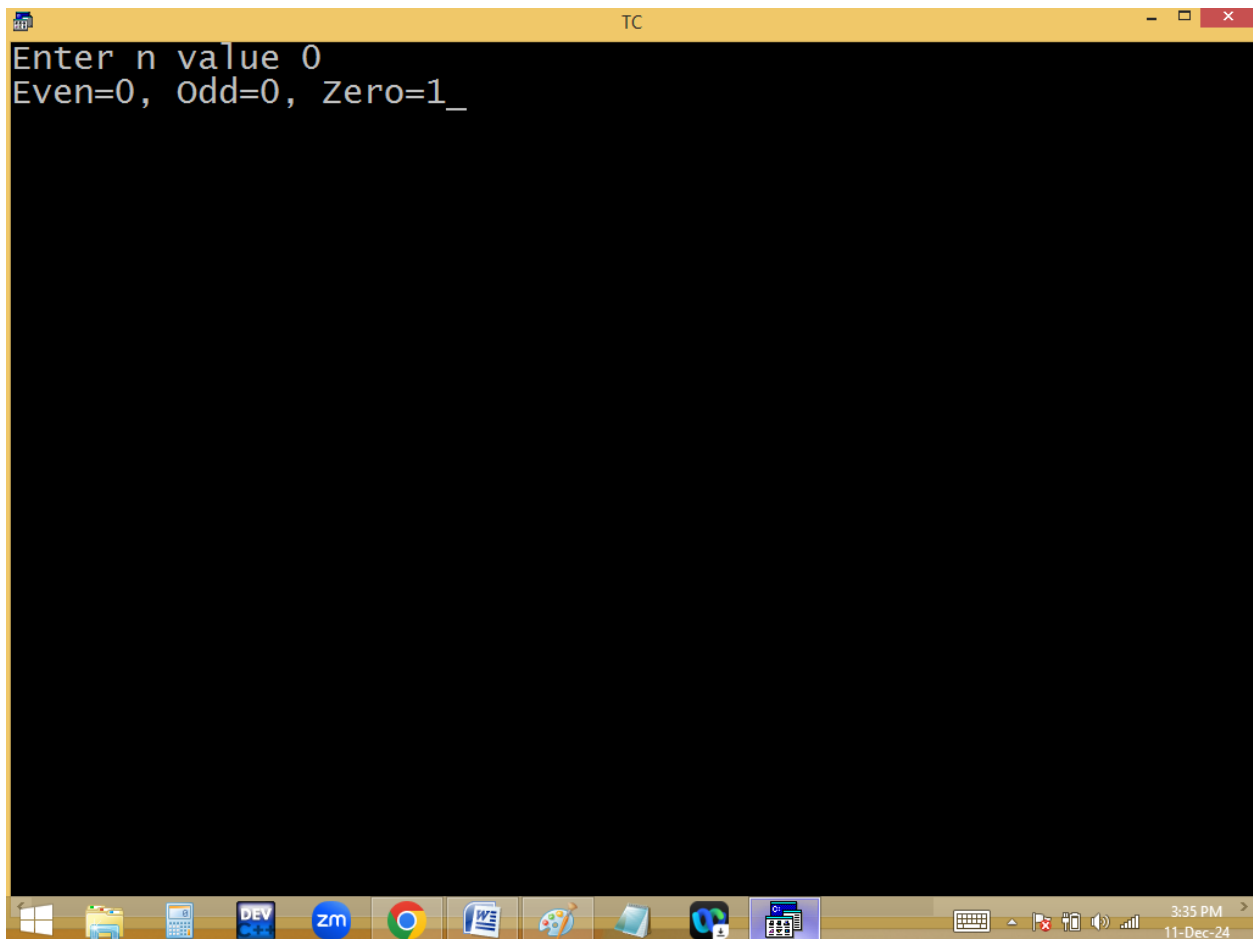




The image shows a screenshot of the Turbo C++ (TC) IDE. The window title is "TC". The menu bar includes File, Edit, Run, Compile, Project, Options, and Debug. The status bar at the top indicates "Line 14 Col 14 Insert Indent Tab Fill Unindent * E". The code is written in yellow text on a blue background. It includes headers for stdio.h and conio.h, and defines a main function. The program prompts the user to enter a number 'n', then uses a while loop to process each digit of 'n'. For each digit, it checks if it is even (increasing 'z'), odd (increasing 'e'), or zero (increasing 'o'). The loop continues until 'n' becomes 0. Finally, it prints the counts of even, odd, and zero digits and waits for a key press before exiting.

```
#include<stdio.h>
#include<conio.h>
void main()
{
long n; int r,e,o,z;
clrscr();
printf("Enter n value "); scanf("%ld",&n);
e=o=z=0;
do
{
r=n%10;
if(r==0)z++;else if(r%2==0)e++;else o++;
n=n/10;
}while(n!=0);
printf("Even=%d, Odd=%d, Zero=%d",e,o,z);
getch();
}
```

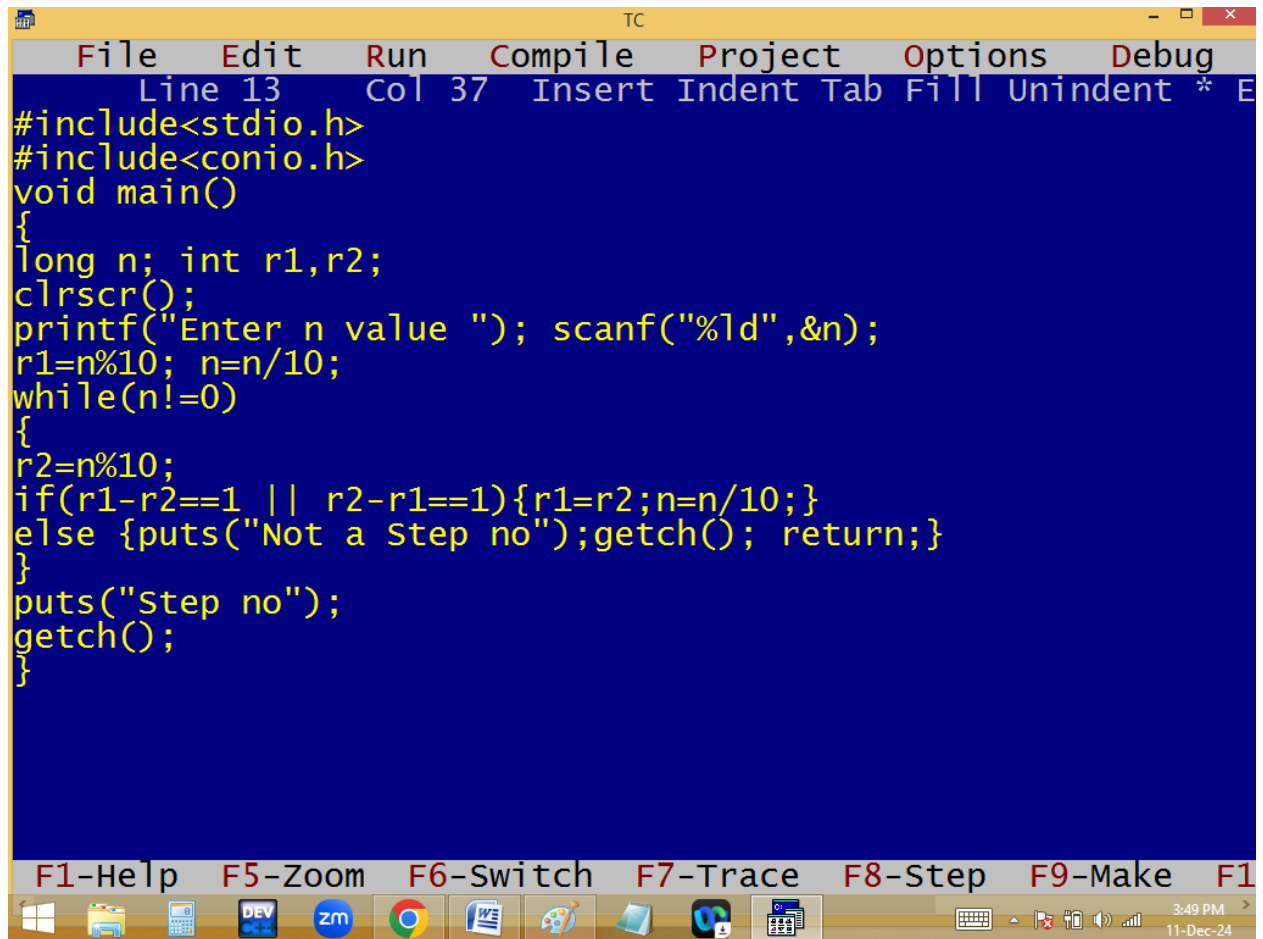
The bottom of the window features a toolbar with function key shortcuts: F1-Help, F5-Zoom, F6-Switch, F7-Trace, F8-Step, F9-Make, and F10-Run. The Windows taskbar at the very bottom shows the Start button, several application icons (including DEV, zm, Chrome, Word, and a game controller), and the system clock indicating 3:35 PM on 11-Dec-24.



Finding step no or not?

1234 → step no

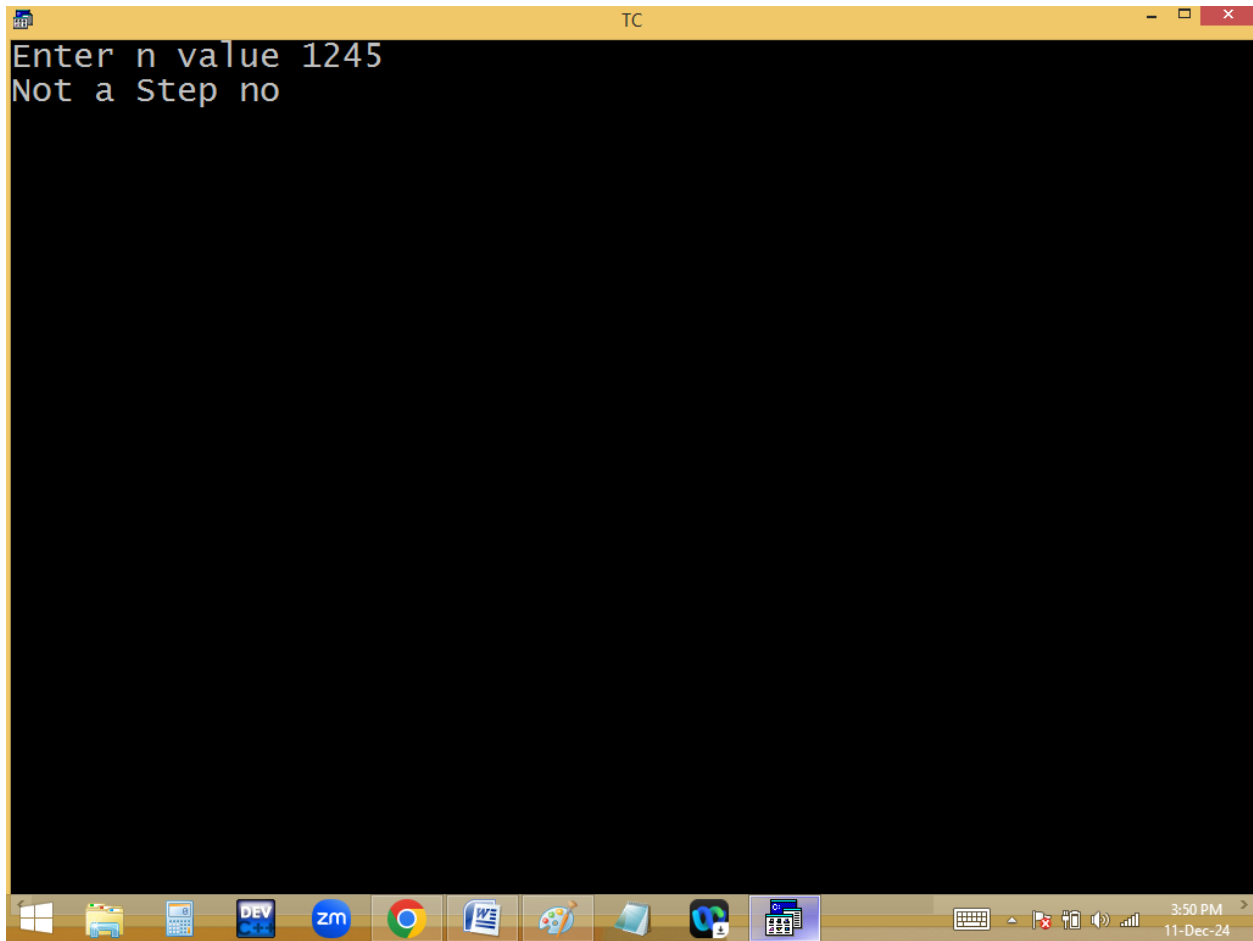
1245 → not a step no

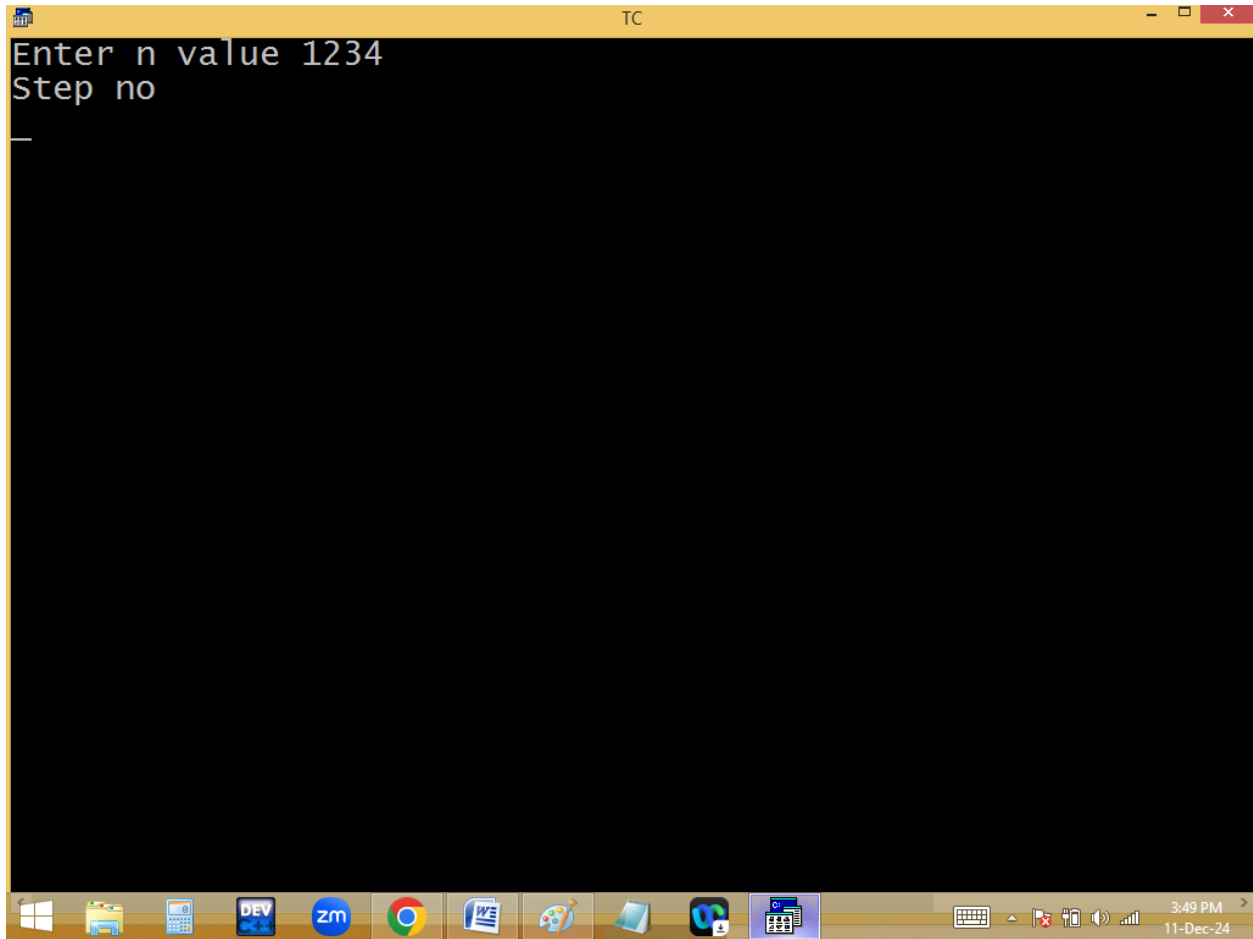


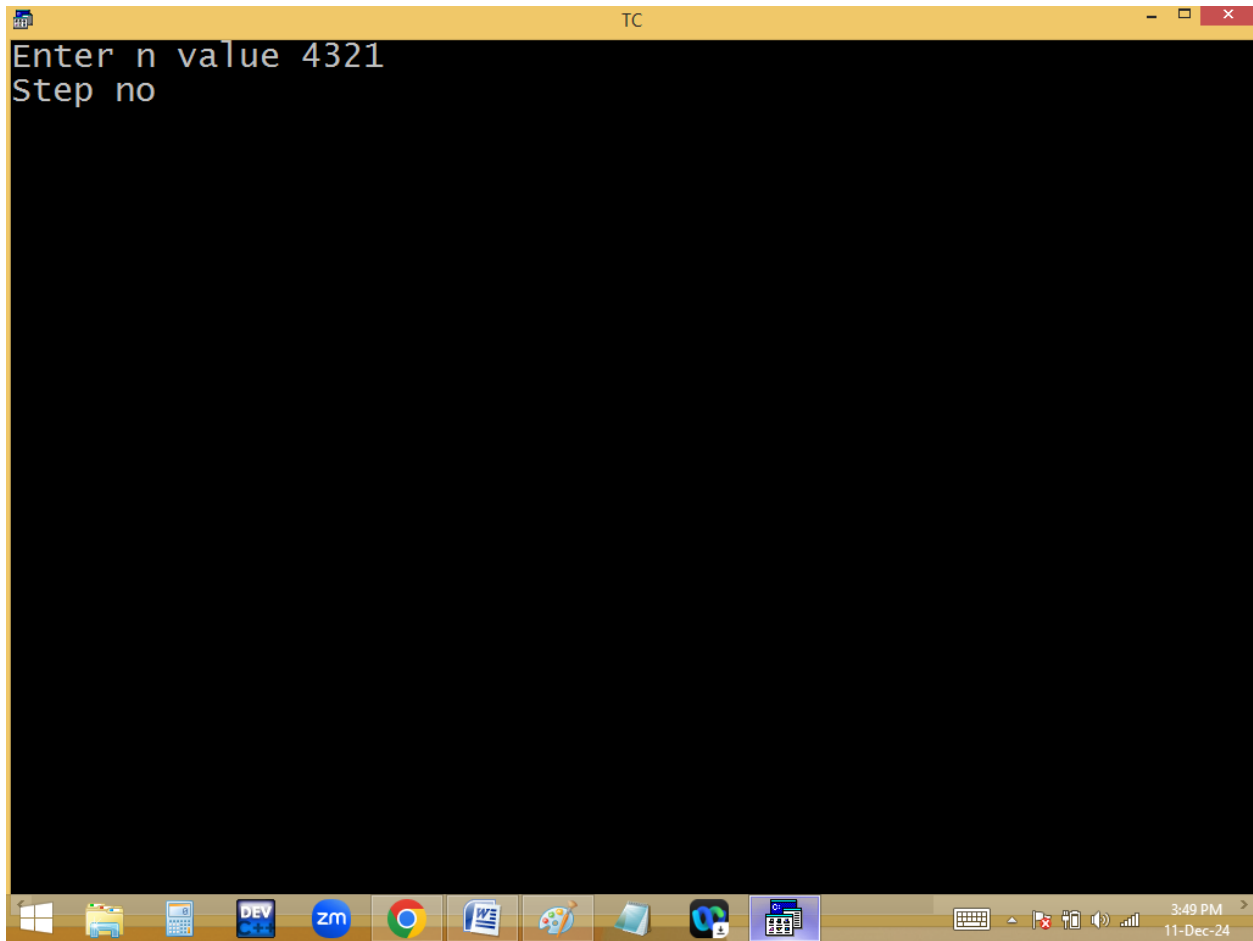
The image shows a screenshot of a Turbo C++ (TC) IDE window. The title bar at the top reads "TC". The menu bar includes "File", "Edit", "Run", "Compile", "Project", "Options", and "Debug". Below the menu bar, the status bar shows "Line 13", "Col 37", and "Insert Indent Tab Fill Unindent * E". The main editing area has a blue background and contains the following C code:

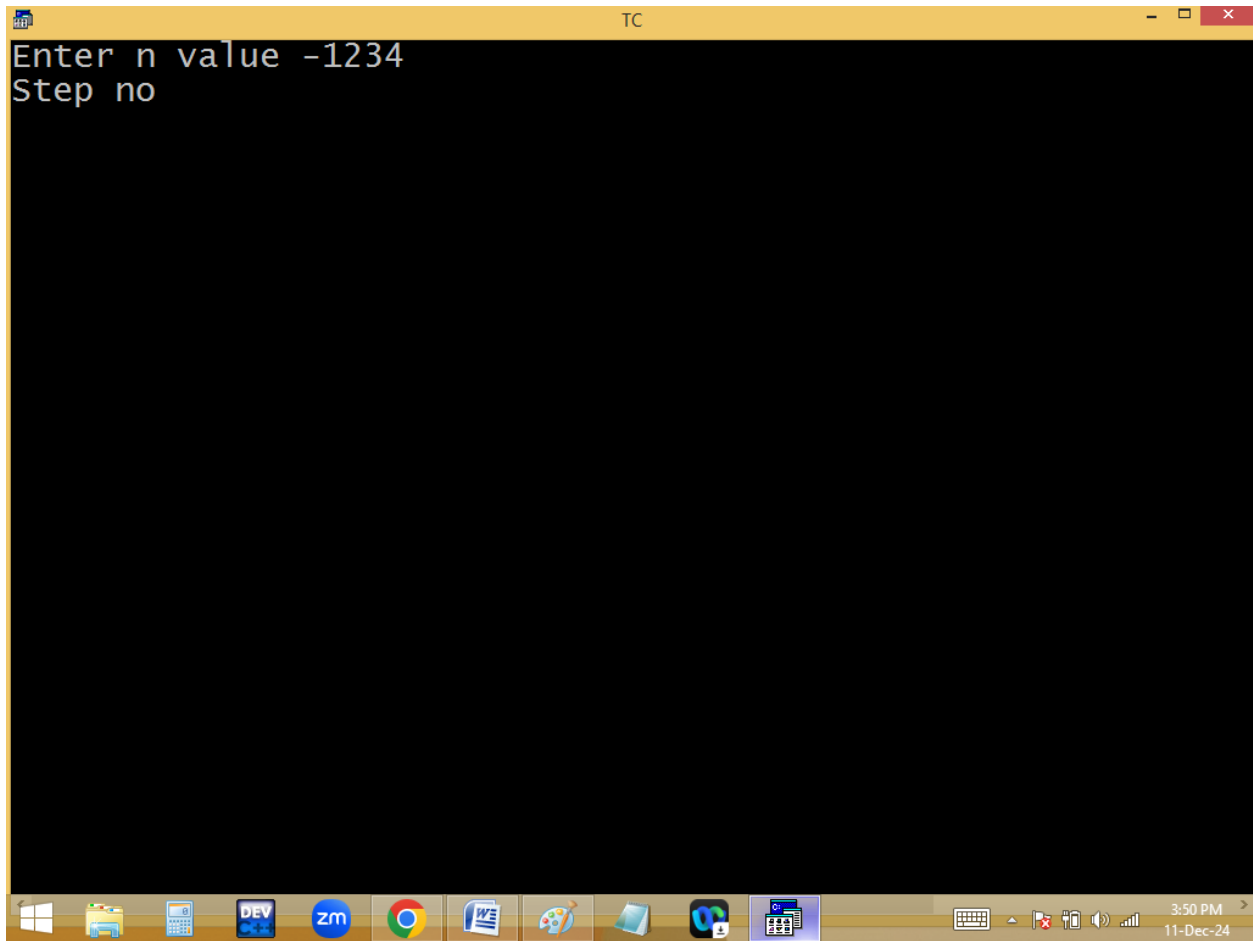
```
#include<stdio.h>
#include<conio.h>
void main()
{
    long n; int r1,r2;
    clrscr();
    printf("Enter n value "); scanf("%ld",&n);
    r1=n%10; n=n/10;
    while(n!=0)
    {
        r2=n%10;
        if(r1-r2==1 || r2-r1==1){r1=r2;n=n/10;}
        else {puts("Not a Step no");getch(); return;}
    }
    puts("Step no");
    getch();
}
```

At the bottom of the window, there is a toolbar with function key shortcuts: "F1-Help", "F5-Zoom", "F6-Switch", "F7-Trace", "F8-Step", "F9-Make", and "F10-Run". The Windows taskbar is visible at the very bottom, showing icons for the Start menu, File Explorer, Calculator, DEV C++, Zoom, Google Chrome, Word, Paint, and other applications. The system clock in the bottom right corner indicates the time is 3:49 PM on 11-Dec-24.









Home work:

1. 12345678 → 78 56 34 12

2. 102 → One Zero Two

